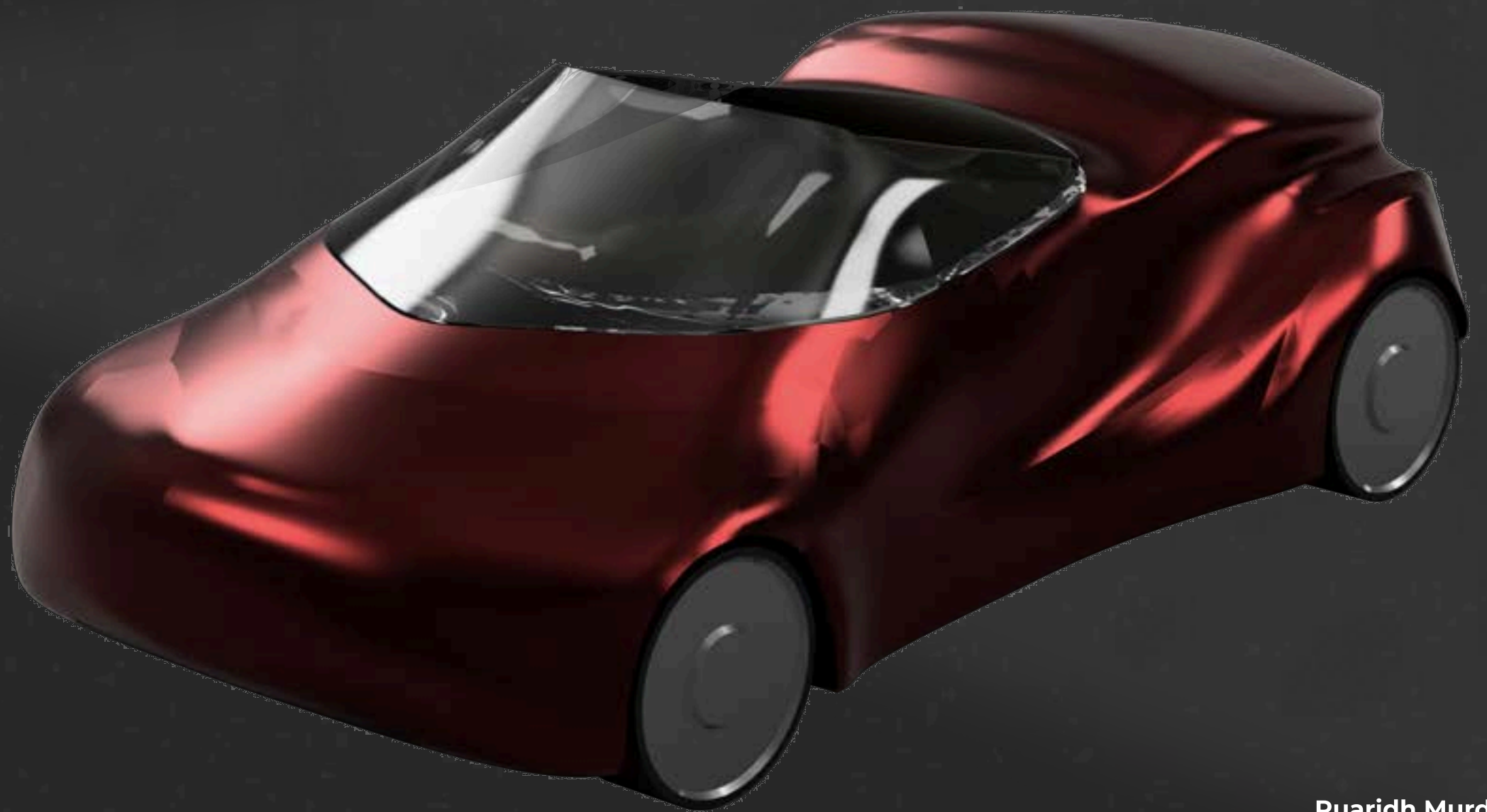


THERMOFLUIDS

Aerodynamic Vehicle Design

- 1. Design Summary*
- 2. Design Rational*
- 3. Benchmarking*
- 4. Conceptual Design Development*
- 5. Final Design*
- 6. CAD Model*
- 7. CFD Setup*
- 8. Meshing & Solution Setup*
- 9. Pressure & Velocity Analysis*
- 10. Streamlines & Turbulence Analysis*
- 11. Aerodynamic Forces*
- 12. Improvements*



This report presents two **thermofluids design studies**: an **open-top four-seater cabriolet** and a **liquid-cooled battery system**. The vehicle project progressed from **concept development** and **CAD modelling** to **CFD analysis** and **refinement**, while the battery study involved **research**, **concept generation**, and **CAD development** of a liquid cooling plate. [20]

Car Overview

The car was designed mainly to **control the rear wake** of an open four-seater cabriolet.

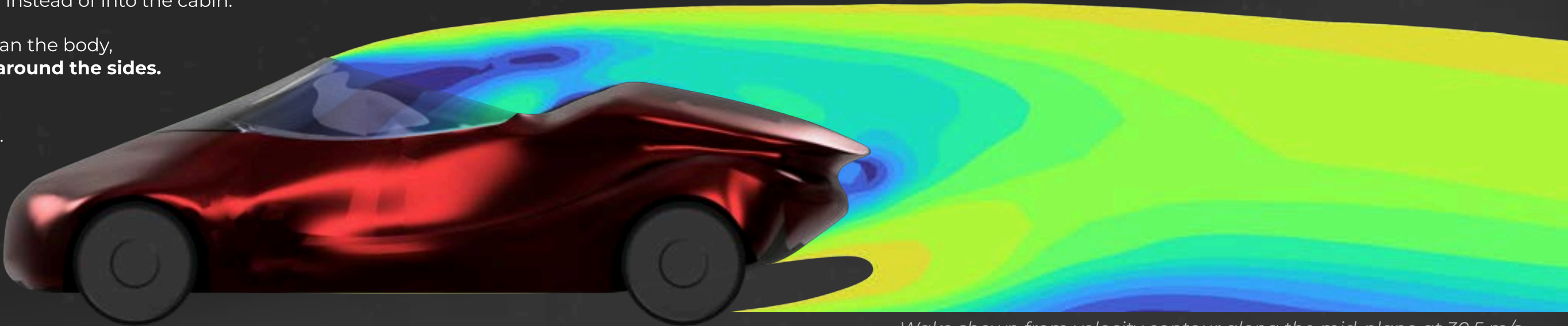
The **front surface** and **curved windshield** guide **airflow around the body** instead of into the cabin.

The **cabin** is **narrower** than the body, encouraging **air to pass around the sides**.

Air channels then **guide** this **flow** toward the rear.

The **extended rear body** allows the air to **slow gradually** and **reattach**, improving pressure recovery and reducing wake size.

Drag Coefficient: 0.458 **Drag Force:** 758N **Lift Coefficient:** 0.135
Lift Force: 223N

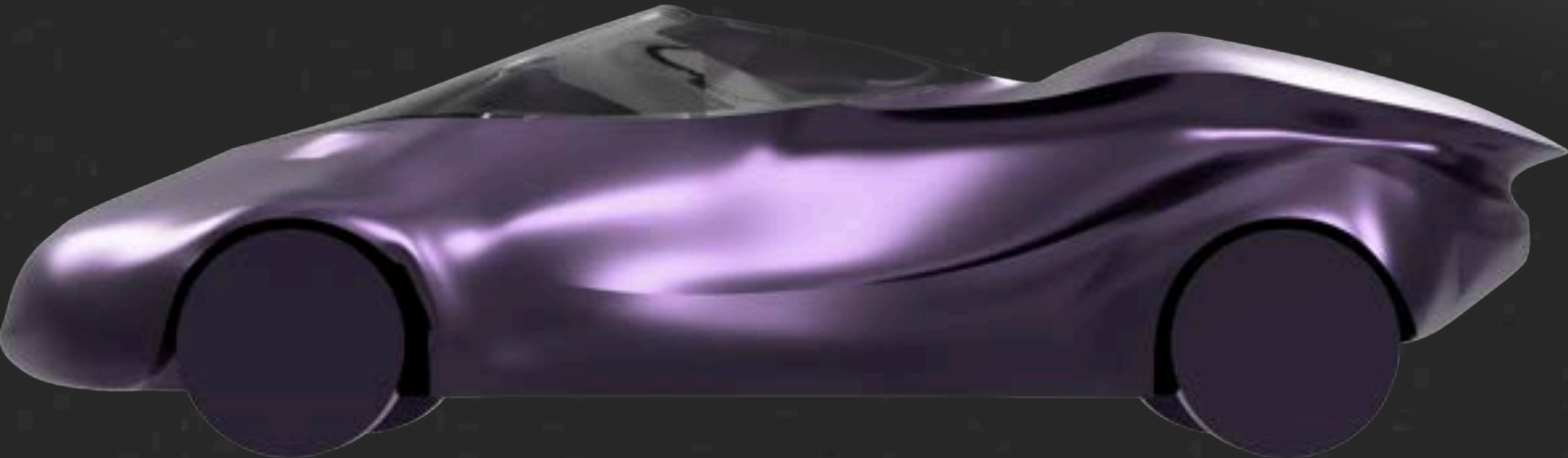


Wake shown from velocity contour along the mid-plane at 30.5 m/s.

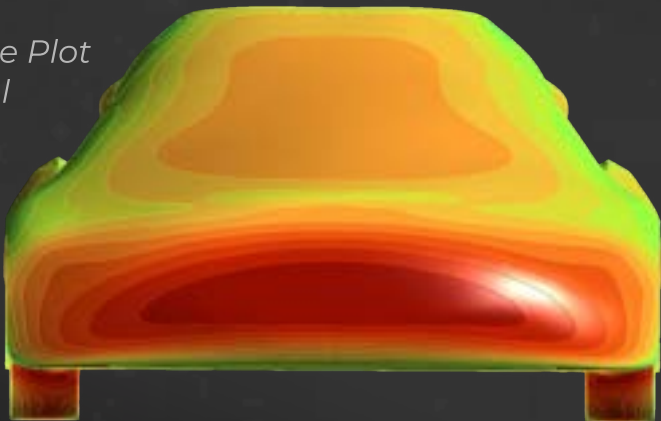
Computational Fluid Dynamics (CFD) simulations conducted in *Ansys Workbench* were used to **analyse** the **velocity**, **pressure**, and **flow behaviour** around the vehicle, allowing **assessment** of the **aerodynamic performance** of the design.

After the initial CFD analysis, **design refinements were implemented** to address high front pressure and side-entry flow. **Updated simulations** showed measurable **improvement**, with **Cd** reduced by **12%** and **drag force reduced** by **14%**.

Improved Design

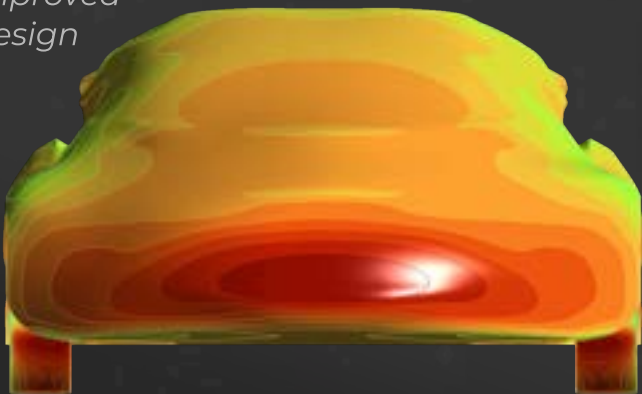


Pressure Plot
Original
Design



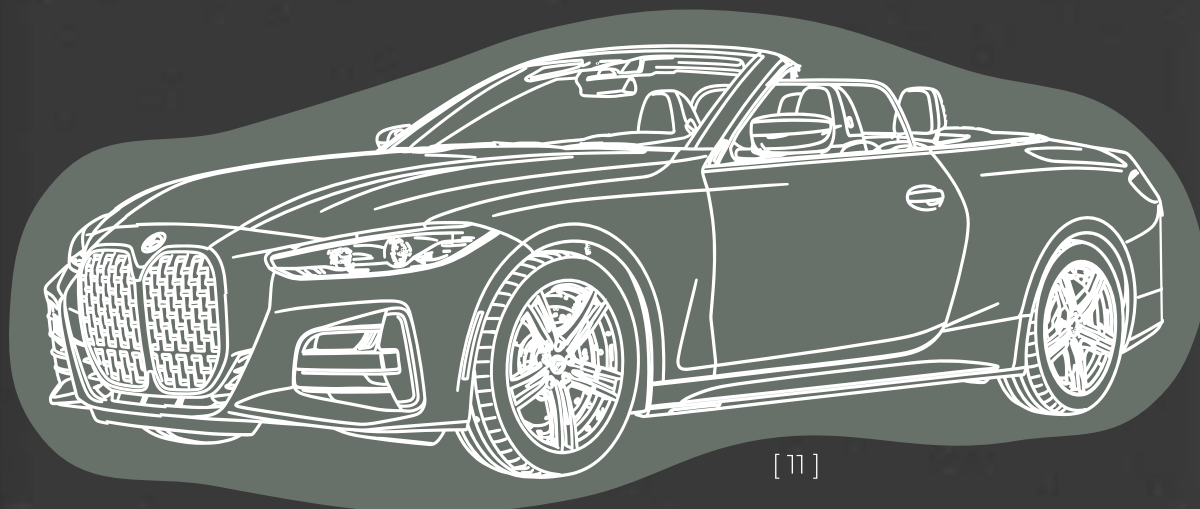
Pressure contours also indicate a smaller and more evenly distributed high-pressure region at the nose.

Pressure Plot
Improved
Design



Vehicle Archetype

A **four-seat cabriolet** intended for everyday road use operated primarily in roof-down conditions.



A four-seat cabriolet was selected due to the **aerodynamic compromise** created by its **open cabin**, where where **aerodynamic efficiency** must be balanced against **passenger comfort** and usability.

Intended User

Comfort-focused roof-down driving at urban and motorway speeds.



User Priorities

- **No wind buffeting** during roof-down driving
- **Low turbulence** within the cabin
- **Low wind noise** at sustained speeds
- **High fuel efficiency** during everyday use

Aerodynamic Implications

- Maintain stable flow at the windshield to avoid wind buffeting
- Avoid strong front-flow deflection, which can increase turbulence within the cabin.
- Maintain attached, stable flow around the cabin entrance to limit pressure fluctuations at high speeds.
- Reduce rear wake size to improve overall aerodynamic efficiency

This **limits aerodynamic intervention** at the front of the vehicle, so the **focus** will be on **controlling how airflow exits** the **cabin** and **forms** at the **back**.

Aerodynamic Characteristics

The **BMW 4 Series Convertible** is used here as an **example** of a modern four-seat production cabriolet, as vehicles in this segment follow **very similar aerodynamic layouts** and **design constraints**.

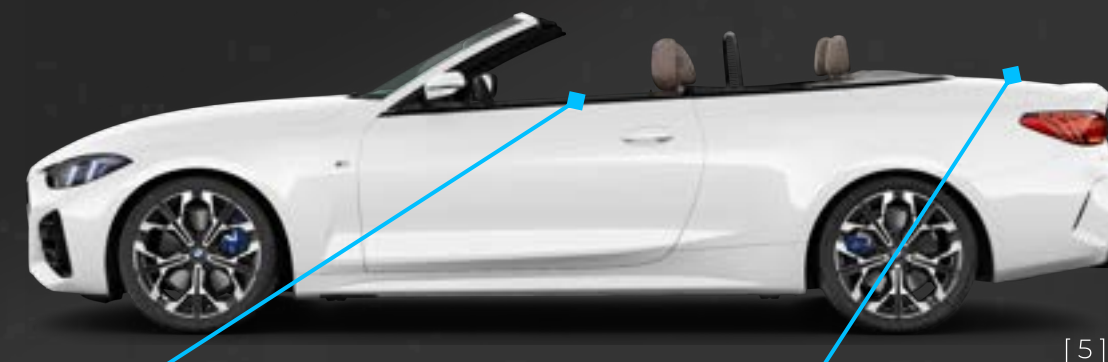
This aerodynamic behaviour is common across mainstream four-seat cabriolets, including the Mercedes-Benz C-Class Cabriolet and Audi A5 Cabriolet. [6][7]

Drag Coefficient:

- Roof Up: 0.28
- Roof Down: 0.33

[2]

Drag increase mainly caused by unmanaged flow through and out of the cabin.



Open cabin:

- Air entering the cabin **circulates before exiting** at the rear, **preventing pressure recovery** and **increasing** the size of the **wake** behind. [2]

Rear deck:

- The **short flat rear** deck is used to meet seat and roof-stowage requirements, providing **limited length** to **guide airflow** smoothly. [3]



CFD of wake behind an open-top convertible.

Windshield header:

- Allows air to enter the open cabin to **avoid unstable flow separation**, but this **increases** the amount of **disturbed flow exiting** at the rear and **enlarges** the **wake**. [4]

Rear termination:

- A **sharp rear cutoff** is used to force a clean separation point, producing a **predictable but large wake** behind the vehicle. [4]

In four-seat cabriolets, airflow is typically allowed to **pass through the open cabin** with minimal shaping after the passenger compartment in order to **prioritise stable and comfortable flow** for occupants. This approach is **common** across mainstream production vehicles. As a result, the **flow exiting the cabin** and **rear deck remains largely unmanaged**, leading to early separation, a **large rear wake**, and **increased aerodynamic drag** in roof-down conditions.

Benchmarking was used to analyse how different vehicle geometries **manage airflow through the cabin** and **control rear wake formation**, informing the development of a rear-focused aerodynamic concept.

Porsche 911 Speedster

Windshield

- Lower windshield **reduces** the **height** of the cabin opening.
- **Less air enters** the cabin in roof-down driving.



[12]

Cabin

- Smaller cabin opening **limits** the amount of **air circulating** into and out of the cabin.

Rear Fairings

- Raised fairings provides a surface for disturbed cabin air to **reattach** after exiting the open cabin.
- This **delays separation** caused by the adverse pressure gradient immediately behind the cabin.
- Rear taper allows **pressure to recover gradually**, shifting separation back and reducing wake size.



[12]

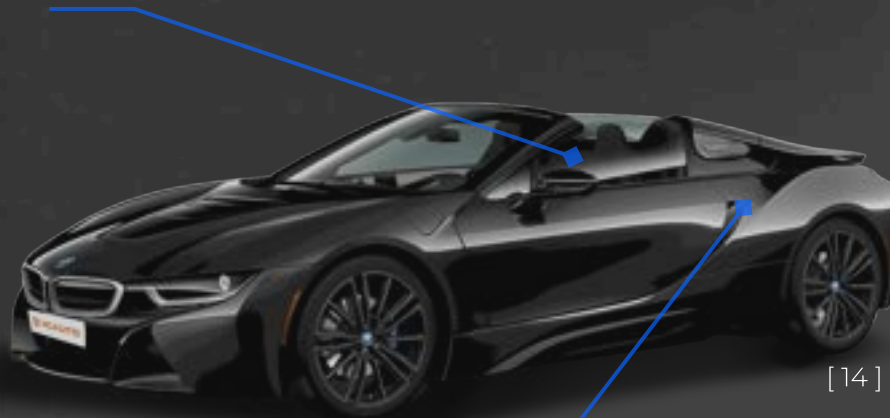
- The rear fairing geometry also accommodates **airflow** delivery to the **rear-mounted engine** intake without disrupting the reattachment process.

[10]

BMW i8 Roadster

Cabin

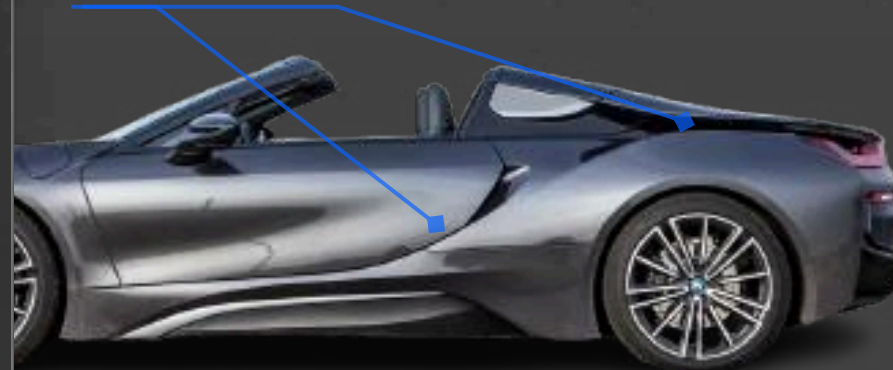
- The cabin opening is **partially enclosed** at the sides.
- This **limits direct interaction** between low-energy, turbulent cabin air and higher-energy external airflow.
- This **reduces pressure fluctuations** at the cabin edge, lowering wind noise. [13]



[14]

Flow Paths

- External airflow is guided around the cabin along defined **side channels**.
- This flow **remains attached** and is directed back toward the **low-pressure wake**[13]



- Supplying higher energy airflow into the wake **increases pressure** behind the vehicle.
- This reduces wake width and **lowers pressure drag**.

Toyota Prius

Body Shape

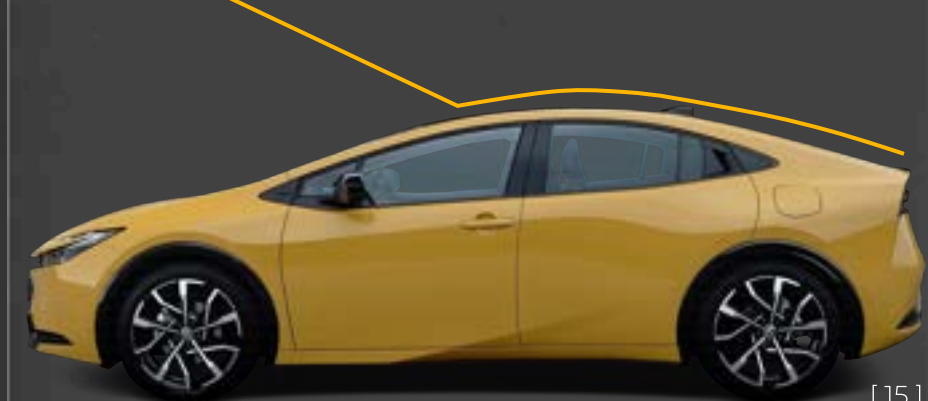
- **Short front section** with a compact hood and **steeply raked windshield**.
- Vehicle proportions place a greater share of the overall **length behind** the cabin.
- This layout supports a **long continuous rear** profile.



[15]

Wake Formation

- The extended rear body follows a **fastback profile** with **gradual taper**.
- This allows **pressure to recover** along the rear surface before separation occurs.
- **Separation is delayed** and occurs in a controlled manner, **reducing wake** and increasing pressure.



[15]

- The **same proportioning principles** can be **adapted to roof-down** vehicles to reduce wake.

[16]

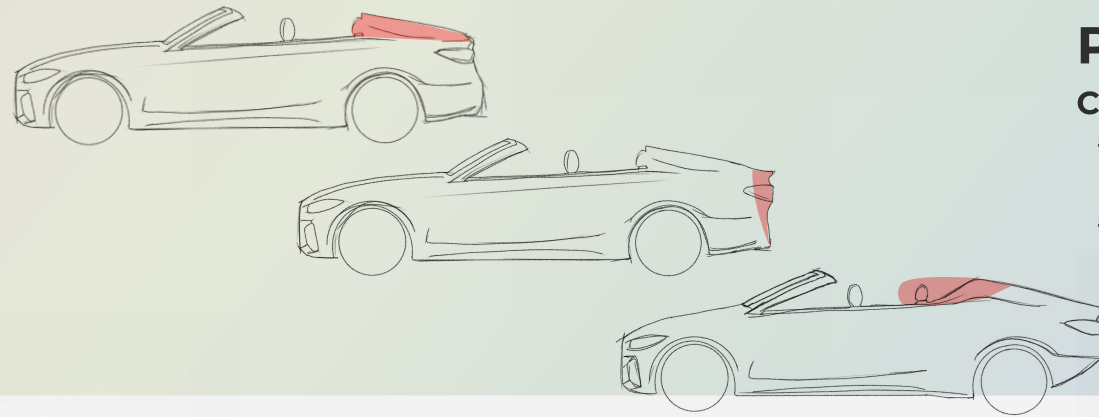
CONCEPTUAL DESIGN DEVELOPMENT

The concept was developed through **progressive changes** to rear geometry, cabin edges, and overall proportions, with a **focus on controlling cabin exit flow and rear wake formation** in roof-down driving.

Phase 1

Rear Shape Development

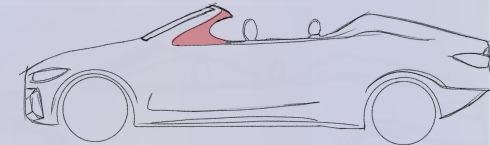
- A **raised rear shape** is added behind the seats to give cabin airflow a surface to **reattach** to.
- The rear is **extended and tapered** to allow **pressure** to **recover** before separation.
- Separation moves **further back** reducing wake size.



Phase 2

Cabin Edge and Side Flow

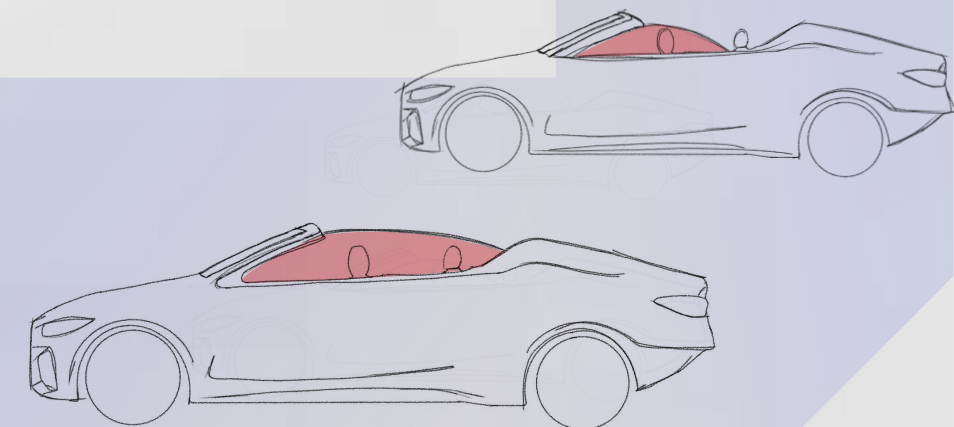
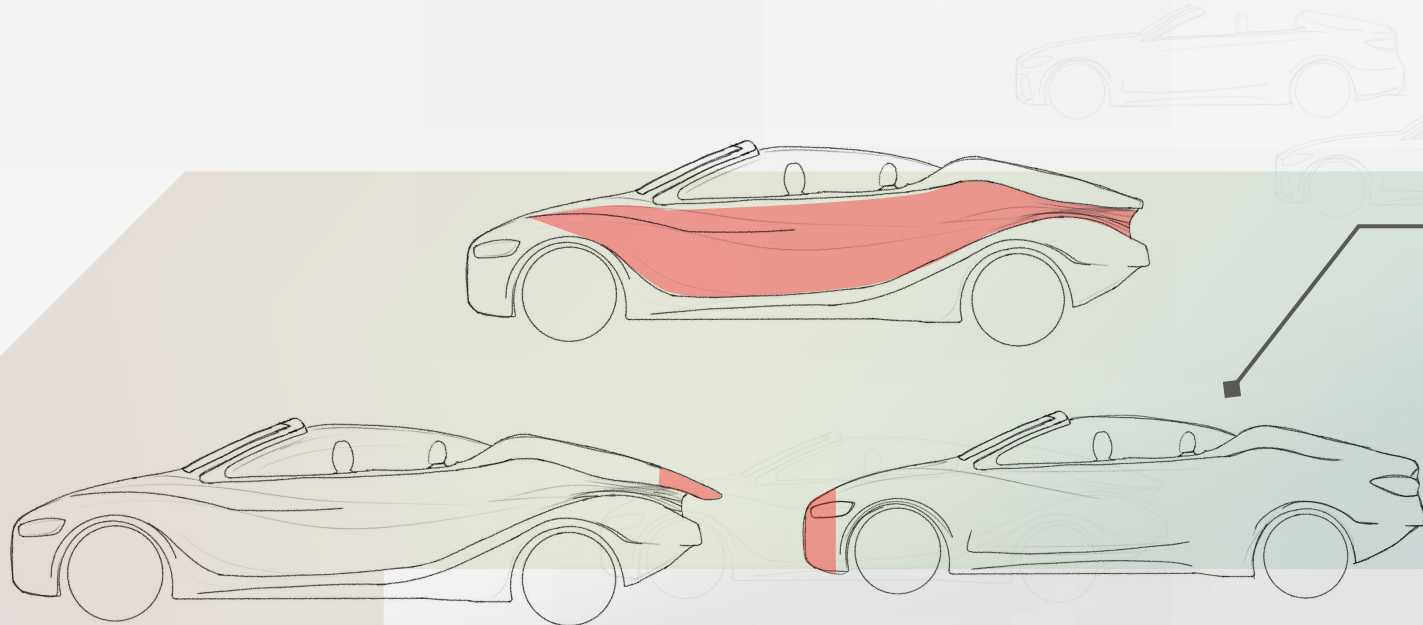
- The cabin opening is **enclosed at the sides** to stabilise the cabin shear layer.
- This **reduces pressure fluctuations** and wind noise.



Phase 3

Side Flow

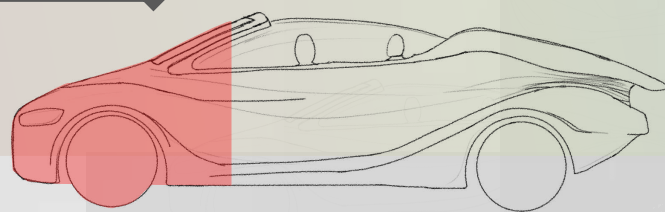
- The concept changes to an **electric vehicle**, removing engine air-intake requirements and allowing a **flat under-tray**.
- External airflow is **guided around the cabin** and directed toward the rear.



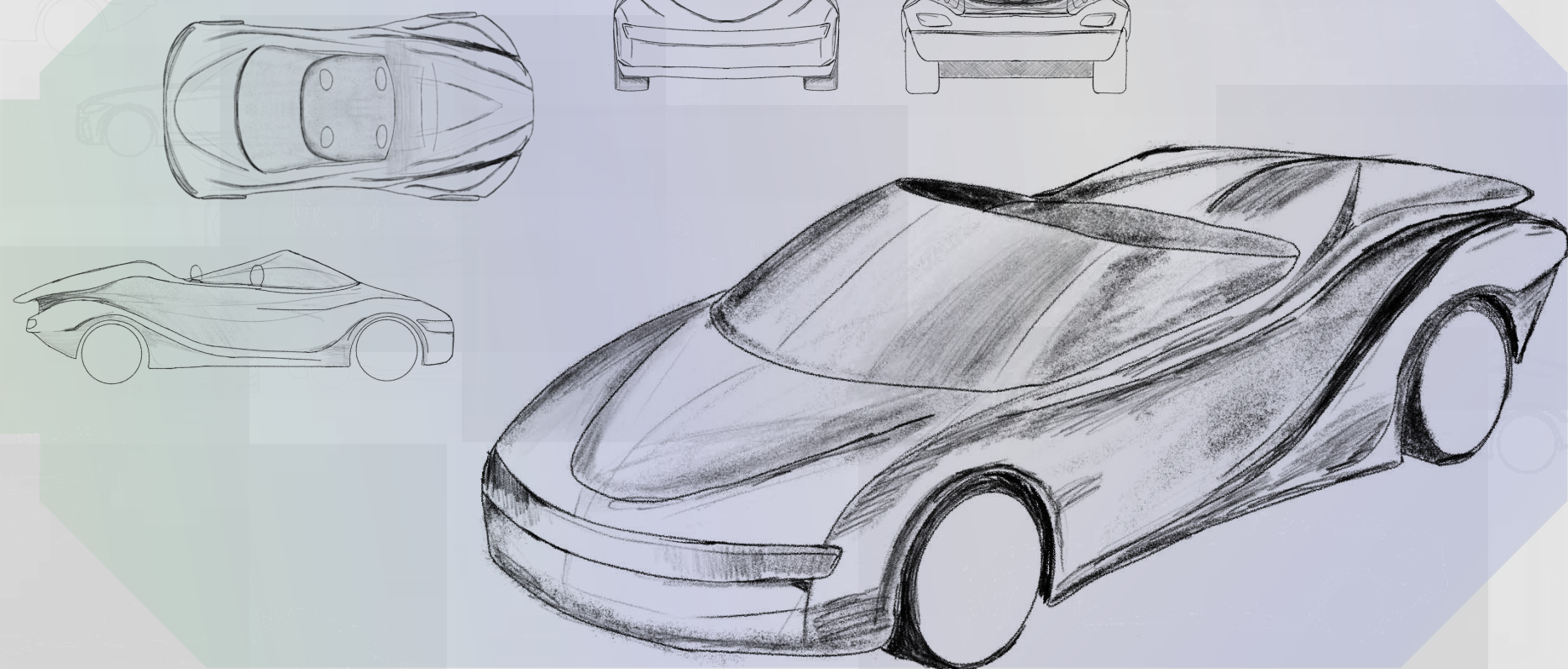
Phase 4

Proportion and Final Geometry

- The front is **shortened and blended** into the windshield to **balance the longer rear**.
- Side flow **channels are deepened** to guide airflow into the wake.



Final Concept



Design Objectives

The aim is to address the key aerodynamic limitations of a four-seat roof-down cabriolet. The primary objectives are to:

- ➖ Reduce overall drag
- ⬆ Improve rear pressure recovery
- ➡ Control wake growth
- 👤 Manage airflow entering/exiting the open cabin

Overall Concept

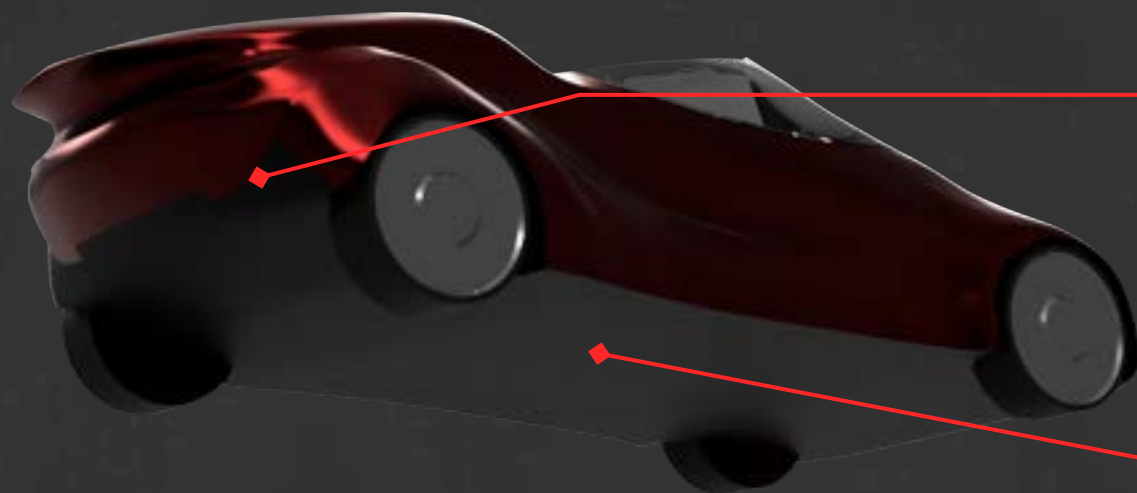
The design focuses on **controlling** how **airflow exits** the **open cabin** and forms the rear wake. The **shape promotes reattachment** and **gradual deceleration** before final separation.

As an electric vehicle, **no front cooling inlets are required**, allowing a **cleaner front surface**. The front and sides guide more air **around the body** instead of into the cabin. The **extended rear body** and **air channels** then manage the rearward flow, aiming to **delay separation**, **improve pressure recovery**, and **reduce the wake**.

Orthogonal Front View

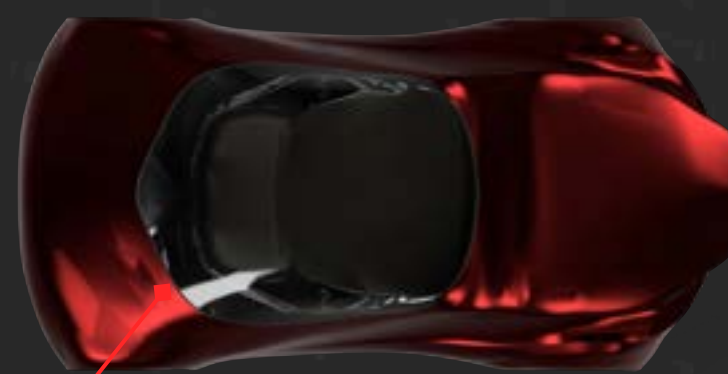


Orthogonal Back View

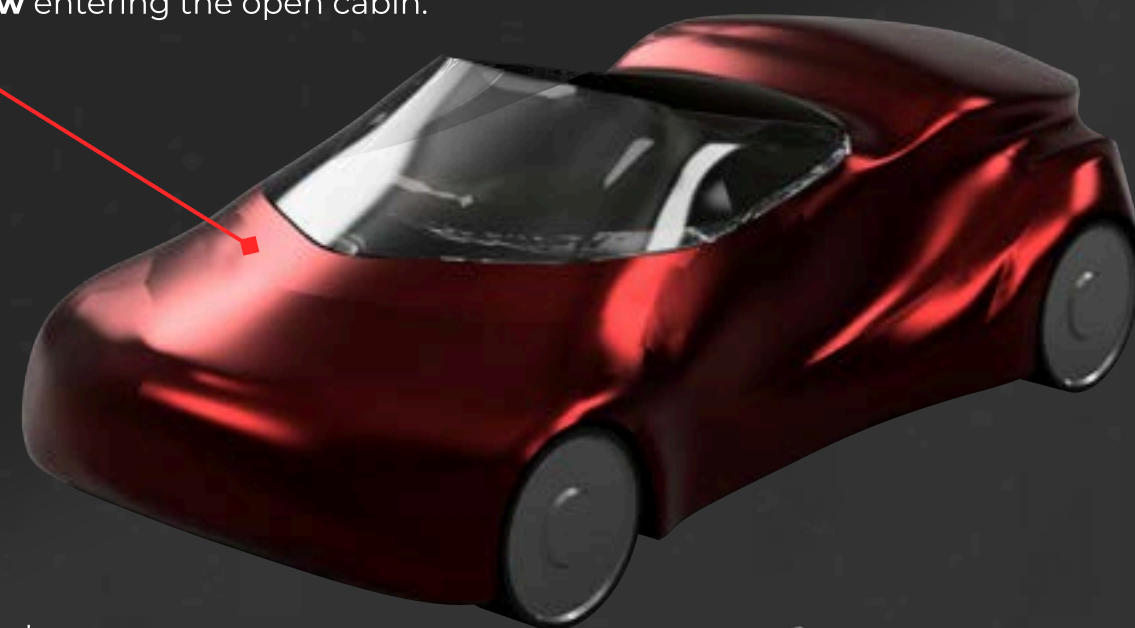


Front Section ➖ 👤

The bonnet transitions **smoothly** into the windshield. This smooths the pressure gradient at the front, **delaying separation**. The **curved** windshield **directs airflow around** the sides **instead of over the top**, **reducing flow** entering the open cabin.

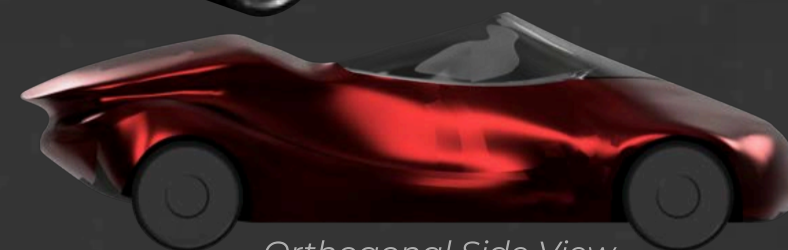


Orthogonal Top View



Cabin Form ➖ 👤

The cabin is **narrower** than the body width and **partially enclosed** at the sides. This encourages **more airflow** to **pass around the body** rather than through the cabin opening, **reducing recirculation**. The **teardrop-inspired** profile promotes **smoother acceleration** and controlled separation.



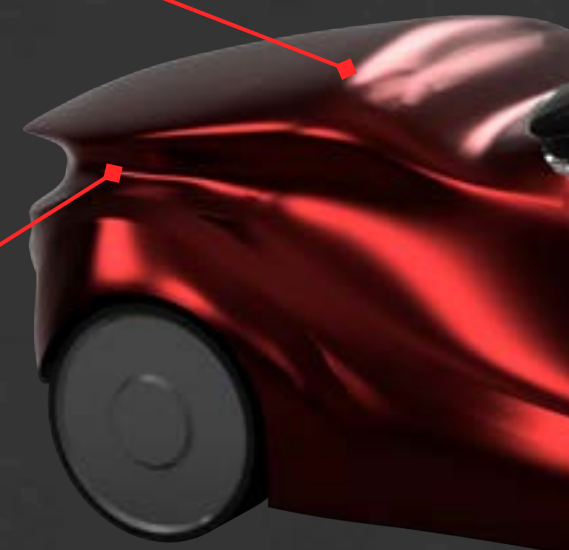
Orthogonal Side View

Rear Hump ⬆ ➡ 👤

The rear body is **lengthened** compared to a typical cabriolet. The raised hump provides a **surface for flow exiting the windshield to reattach** after the cabin opening. By **delaying separation** and allowing **gradual deceleration**, rear pressure recovery improves and **wake size is reduced**.

Air Channels ➖ ⬆ ➡

The side surfaces **curve inward** into defined rear air channels. These channels **guide side flow** toward the trailing edges and **concentrate it near the rear**. Supplying faster, organised flow to the rear helps keep the **wake narrower** and limits wake growth.



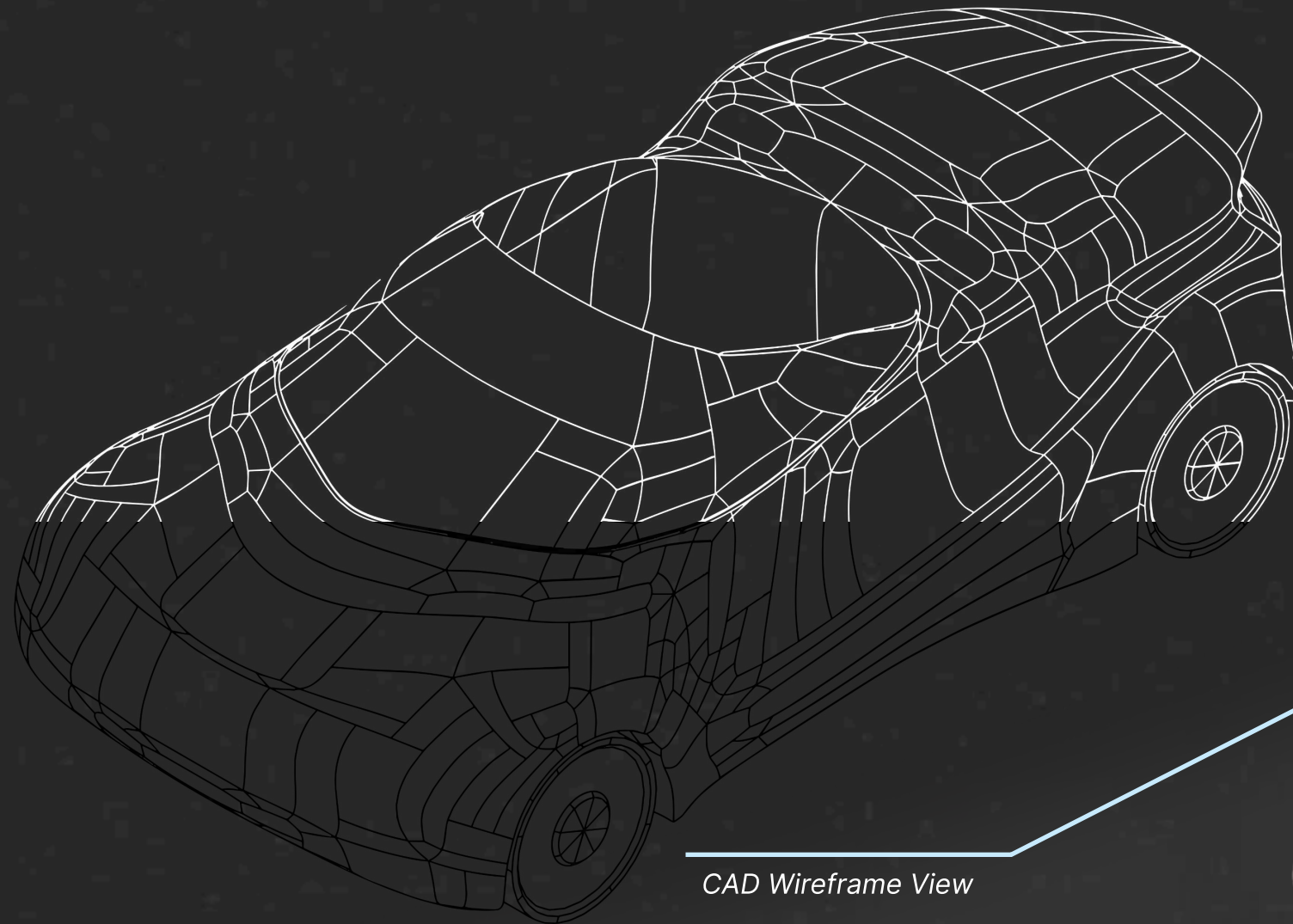
Rear Underside Slope ⬆ ➡

This gradual expansion acts as a **mild diffuser**, allowing underbody flow to expand in a controlled way and helping **limit wake growth** from below.

Flat Undertray ➖ ➡

As an electric vehicle, **no engine cooling inlets are required**, allowing a **smooth front surface**. The electric drivetrain also enables a **fully flat undertray**. A flat undertray **reduces underbody disturbance** and supports **smoother flow beneath the vehicle**.

CAD MODEL



Frontal Area

The model was designed at **1:17** scale. The frontal area was measured by **projecting** the car onto a sketch on the **YZ** plane and inspecting it. The **measured area** was: $10050.162mm^2$

Calculating the frontal area if it were scaled up to real size:

$$17^2 = 289$$

$$10050.162 \times 289 = 2,904,496.818mm^2$$

$$\frac{2,904,496.818}{1,000,000} = 2.904m^2$$



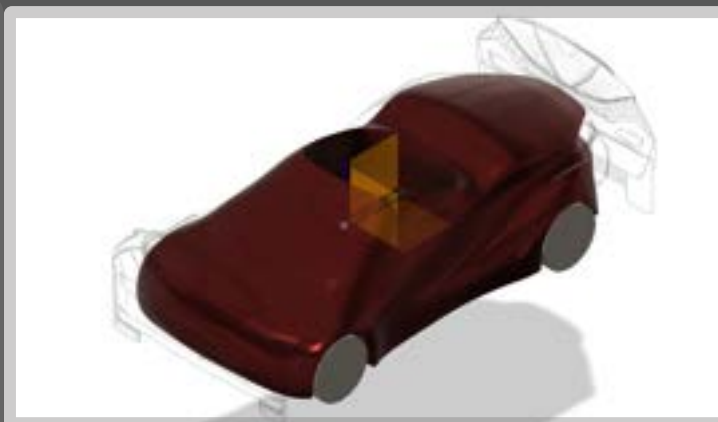
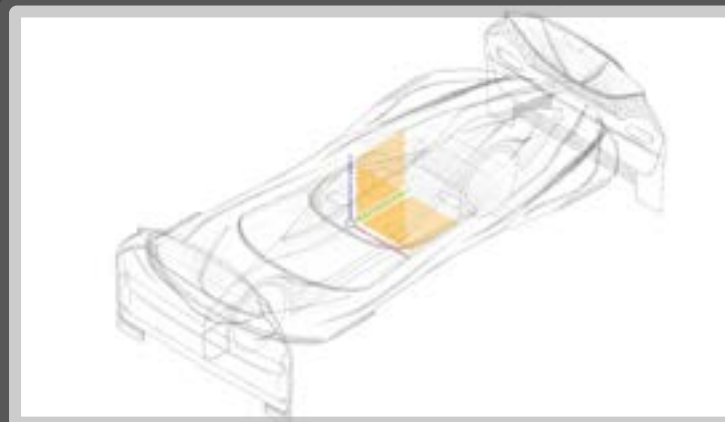
Process

The model was created in **Fusion 360** using the Form workspace. Four **orthographic** reference sketches: *top, side, front and back*, were produced to define the overall proportions and main body lines. The model was produced at **1:17** scale.

The car was modelled with **mirror** enabled to **ensure symmetry**. Sharper features, such as the wheel arches, were defined using the **crease tool**. The underside was made completely flat using a **construction plane** and the flatten command.

Once the main geometry was complete, additional features were added purely **for visuals**. The **windshield was separated** from the body using the **split body tool** and a curved plane to allow different materials to be applied in rendering.

Minor visual refinements were then added to the wheels to improve the look in rendered views. These extra details were **not included** in the CFD version of the model in order to **preserve clean, simplified geometry** for analysis.



This slide shows the **start of the CFD process**. It explains how the **geometry was prepared** and how the surrounding air domain was created to allow accurate aerodynamic simulation. [20] [24]

Preparing

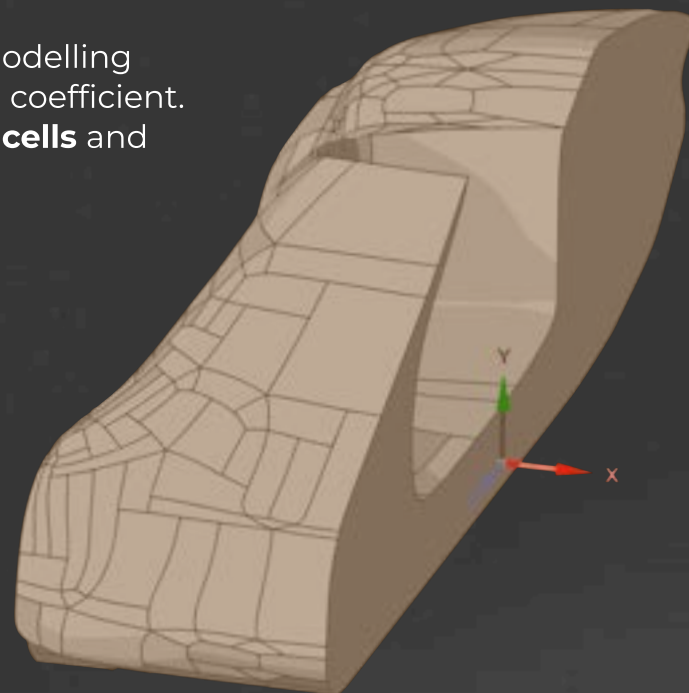
The car was **cut in half** along the centreline using a **construction plane** and the **Split Body** tool. One side was then deleted.

Because the car is **symmetrical**, modelling only half does not change the drag coefficient. This **reduces** the number of **mesh cells** and **shortens simulation time**.

The wheels were modelled as **part of the main solid body**, so no additional operations were required to combine separate components.

Model Scale: 1:17
Length: 283mm
Height: 90mm
Full Width: 130mm

Halved Frontal Area: 5025mm

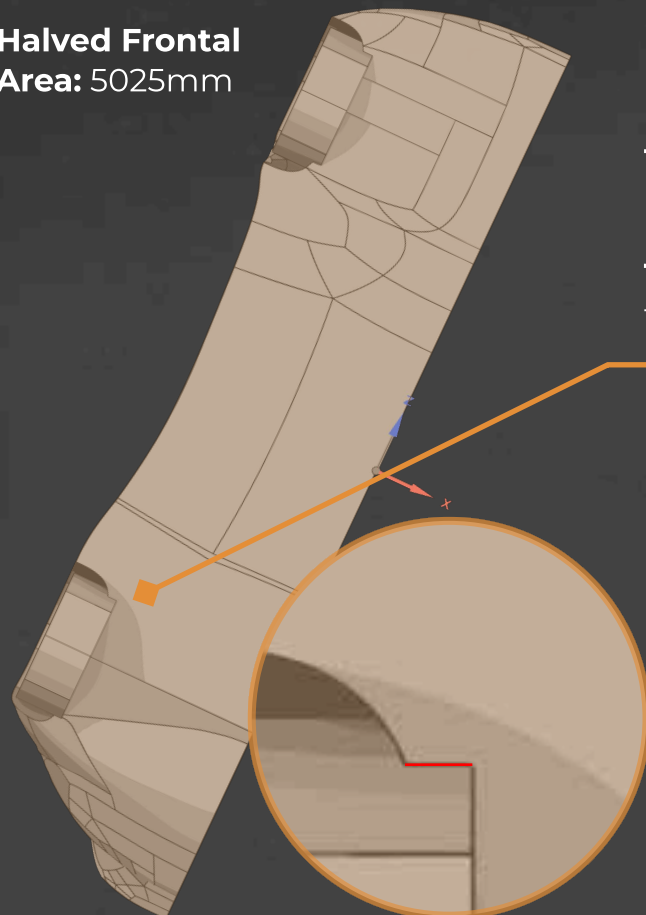


Using the prepare tools on SpaceClaim **four short edges** were identified on the model.

Three were **successfully resolved** using the automatic repair tools in SpaceClaim.

The remaining short edge could not be corrected automatically, so **adjacent faces** were **manually merged** to remove it.

Following these corrections, the geometry check confirmed that **no remaining errors** were present and the model was **fully watertight**.



Enclosure

A surrounding fluid domain must be created to **represent the air** through which the car moves.

Enclosure defined using vehicle dimensions:

- **Front:** 2L
- **Rear:** 5L
- **Height:** 3H
- **Width:** 3W
- **Bottom clearance:** 0.16 mm

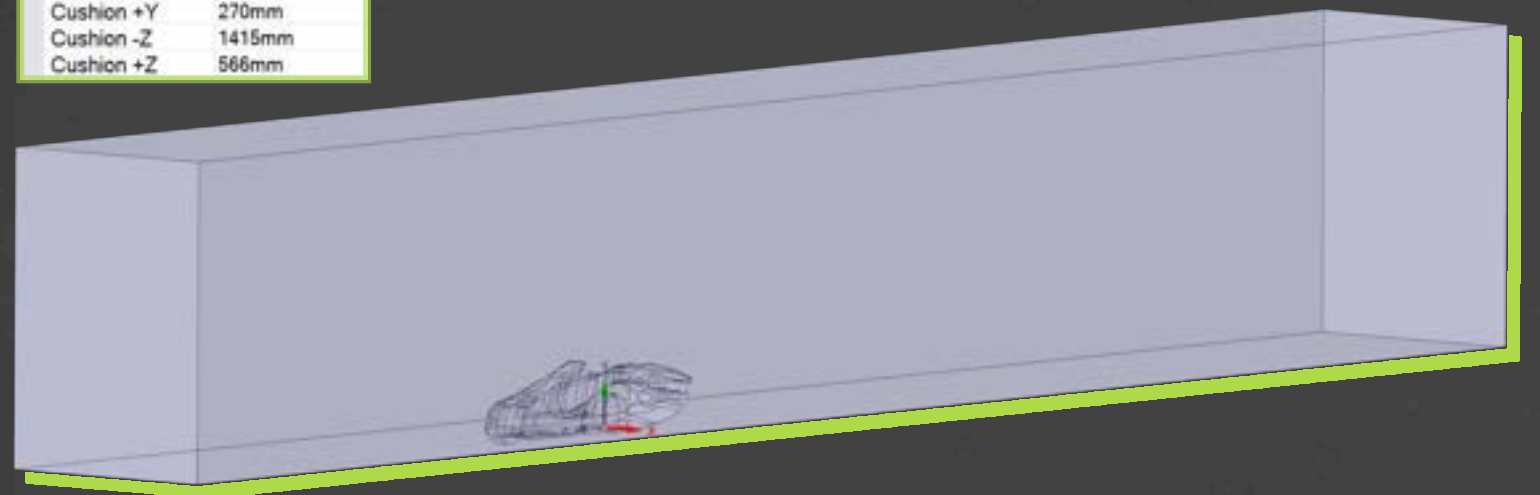
The inlet must be placed far enough from the vehicle to **prevent distortion of the incoming flow**.

This is **enough distance** for the flow to **stabilise** before interacting with the front of the vehicle.

The **wake** region behind the car **contains separation, recirculation and pressure recovery**.

The **5L** distance for the outlet distance ensures a **full wake development, stable pressure** outlet behaviour, and **accurate drag** prediction

Enclosure	
Type	Box
Cushion -X	195mm
Cushion +X	0
Cushion -Y	0.16mm
Cushion +Y	270mm
Cushion -Z	1415mm
Cushion +Z	566mm



The side and top boundaries **must not artificially constrain the flow**. If the enclosure is too small, **blockage** effects **increase** and the velocity field becomes **artificially accelerated**.

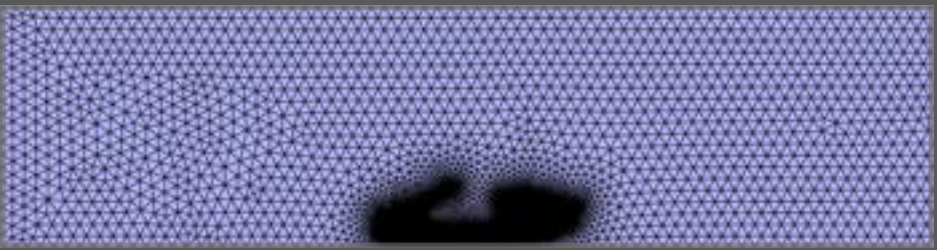
$$Blockage = \frac{5025}{96690} \approx 5\%$$

The frontal area of the car is **less than 5%** of the enclosure **cross-sectional area**, keeping blockage within acceptable limits.



Meshing

After importing the geometry into the Mesh module, the enclosure was treated as a **single fluid region** with **no internal voids**.



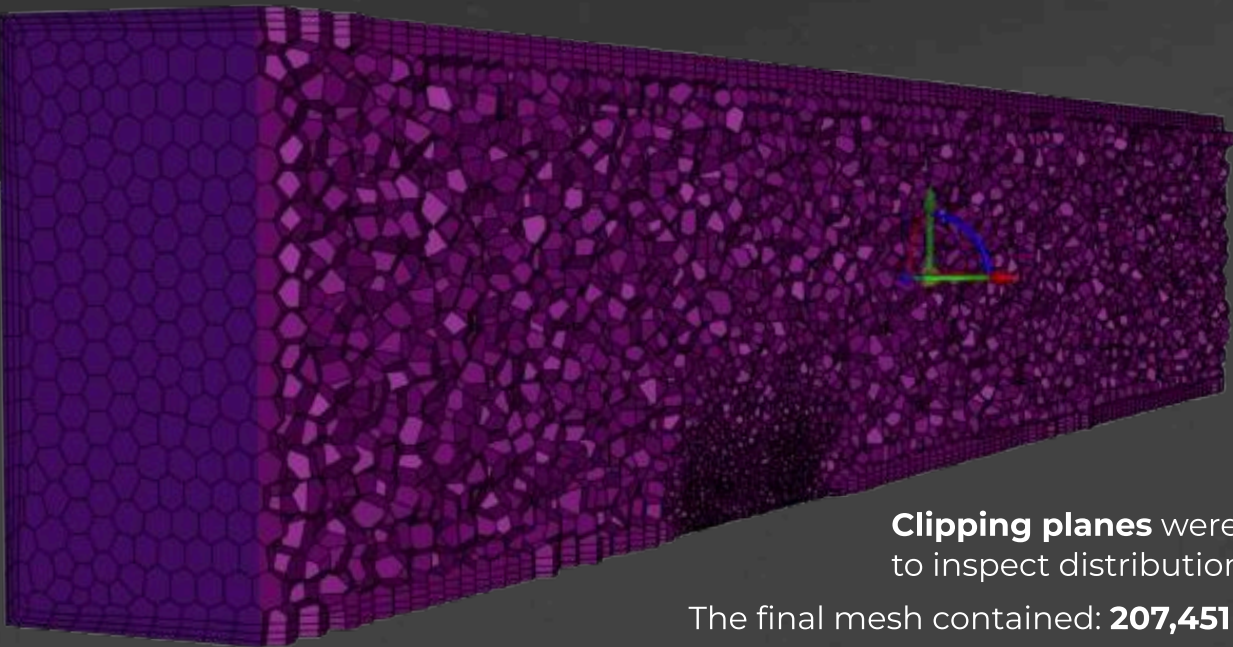
The surface mesh was [20] generated using:

- Minimum base size: **0.001 m**
- Maximum base size: **0.02 m**

A target mesh size of **0.001 m** was set on the **vehicle surfaces**. The enclosure was meshed with cell size **increasing further away from the car**.

This **concentrates detail** where the flow changes a lot, while keeping the overall **number of cells** and computation time **reasonable**.

The domain was described as consisting of **fluid regions only**, since the enclosure is all air. The final volume mesh was then generated with a maximum cell length of **0.02 m**.

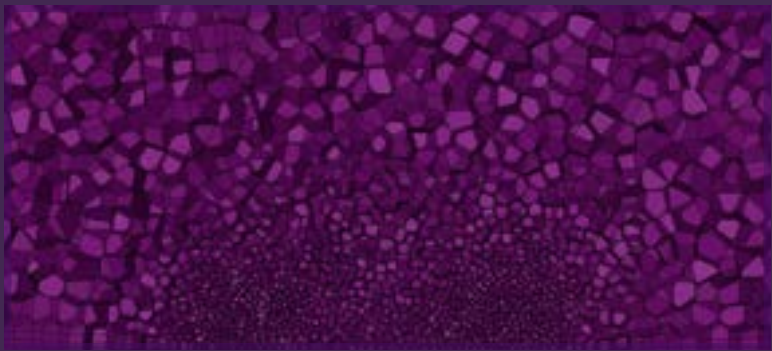


Clipping planes were used to inspect distribution.

The final mesh contained: **207,451 cells**

The minimum orthogonal quality was: **0.11**

Values around **0.1 or higher** are generally considered acceptable, as they **indicate better-shaped cells** and help produce **more accurate** and reliable results.



Solutions Setup

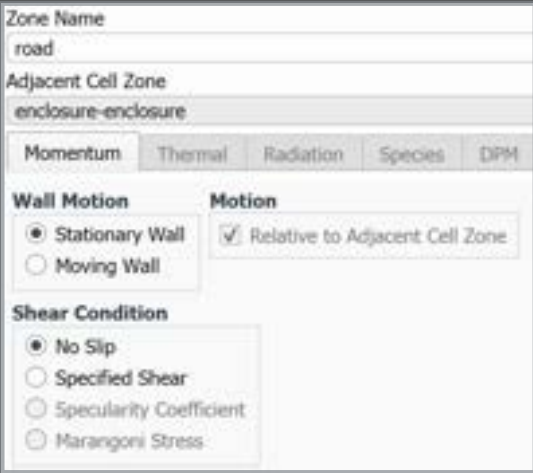
After meshing, the model was opened in Fluent to define the **physical models**, **boundary conditions** and **solver settings**.

A standard **k-ε two-equation** turbulence model was used to **represent turbulent flow behaviour** around the vehicle. External vehicle aerodynamics at this Reynolds number are **fully turbulent**, and the k-ε model provides a **stable and reliable prediction** of separation and wake formation while remaining computationally efficient.

The enclosure region was assigned as **air**, with a standard **density of 1.225 kg/m³** and a **dynamic viscosity of 1.7894 × 10⁻⁵ kg/(m*s)**.

The boundary conditions were applied: [20]

- **Inlet:** Velocity inlet (30.5m/s)
- **Outlet:** Pressure outlet (0 Pa gauge)
- **Car surface:** No-slip wall
- **Road surface:** No-slip wall
- **Symmetry plane:** Symmetry condition



The road was defined as a **stationary wall** to replicate **wind tunnel conditions**. In wind tunnel testing, the vehicle remains fixed to the floor of the tunnel.

The **30.5m/s** corresponds to **68 mph**, which is a typical motorway driving condition.

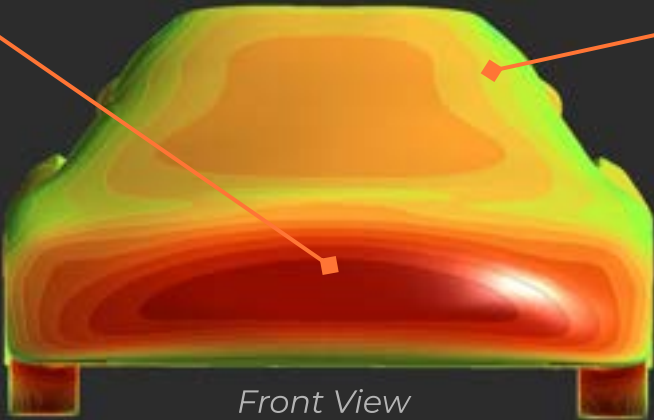
Reference values were computed from the car surface to ensure **correct coefficient calculation**.

Reference Values	
Area [m²]	0.005025081
Density [kg/m³]	1.225
Enthalpy [J/kg]	0
Length [m]	0.283
Pressure [Pa]	0
Temperature [K]	288.16
Velocity [m/s]	30.523
Viscosity [kg/(m s)]	1.7894e-05
Ratio of Specific Heats	1.4
Yplus for Heat Tran. Coef.	300

The simulation was run for up to **500 iterations** or **until stable behaviour was observed**.

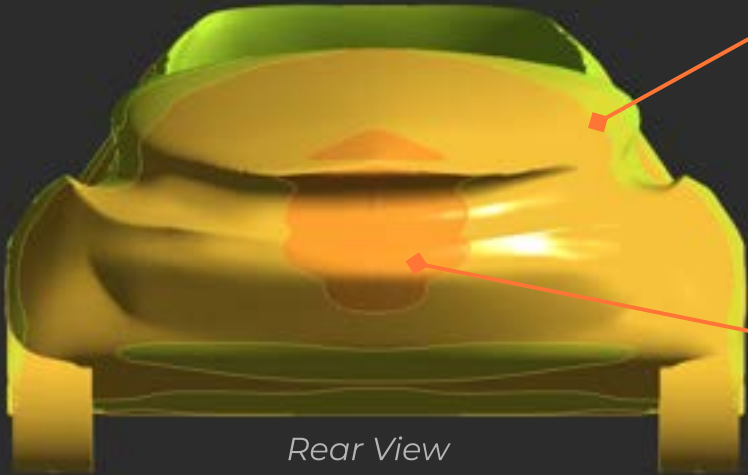
2D and 3D Pressure Plots

A strong **high-pressure** region forms at the **nose**. This stagnation pressure is the highest on the vehicle and **significantly contributes** to drag.



Front View

Pressure **decreases smoothly** over the bonnet and **around the side, maintaining attached flow** and preventing early separation.



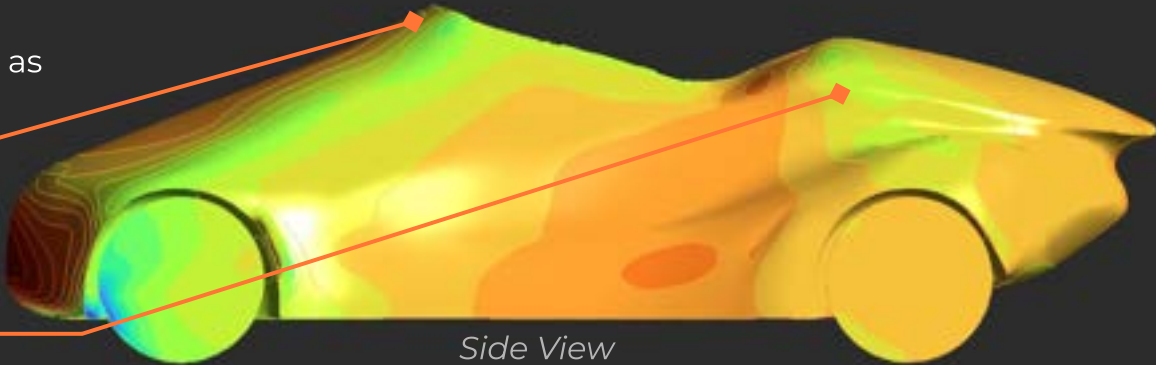
Rear View

Shows **mostly moderate pressure, without a strong low-pressure suction zone**. Pressure **transitions gradually** across the upper and lower rear surfaces, with **no sharp drop** at the edge.

The rear **pressure deficit** is b, so the pressure **recovery** is likely **improved** relative to a short flat rear deck.

Pressure **drops over the windshield** as the flow accelerates into the open cabin.

Pressure is **fairly uniform** which means the **flow stays attached** as it moves toward the rear hump.



Side View

Mid-Plane View

Inside the cabin, **recirculating flow** creates a **low-pressure region**.

A **wake** region forms **behind the vehicle**, but the rear pressure drop is limited.

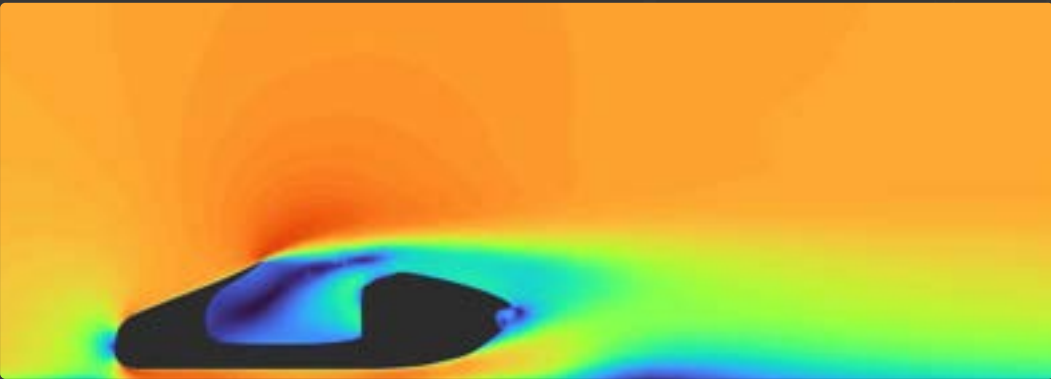
The largest pressure difference in the flow field occurs at the **front stagnation region** rather than the rear.



2D Velocity Plot



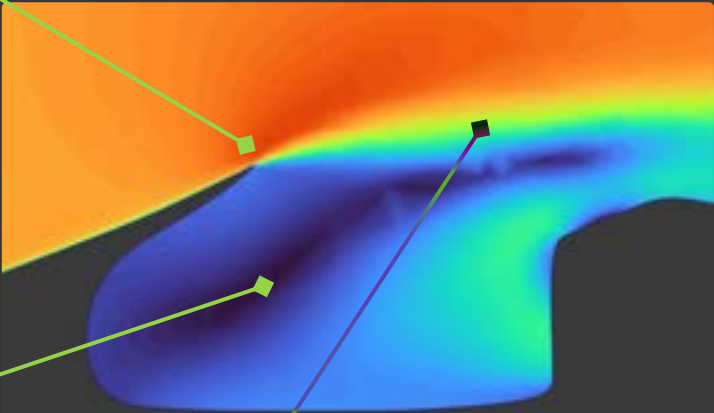
Velocity contours shown along the mid-plane at 30.5 m/s.



Windshield Header

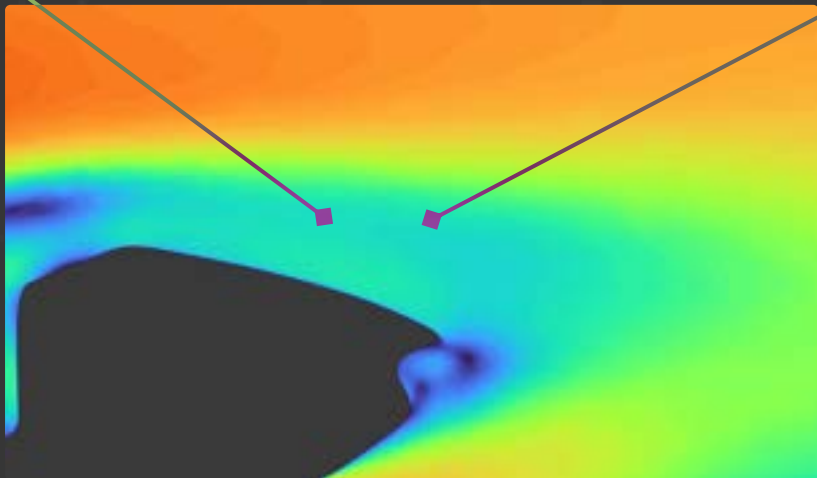
The flow **separates at the windshield header** and enters the cabin, forming a **large recirculation zone**.

This **low-velocity** region corresponds to a pressure deficit inside the cabin and **limits pressure recovery** before the rear, increasing drag.



The rear hump **reduces the cabin's influence on wake growth** but does not remove it.

Above the cabin, the **shear layer** remains **relatively coherent** as it **travels back** rather than breaking down immediately. This allows **higher-momentum flow to reach the rear section** before separation.



The **flow continues** over the rear hump and **stays organised for longer**.

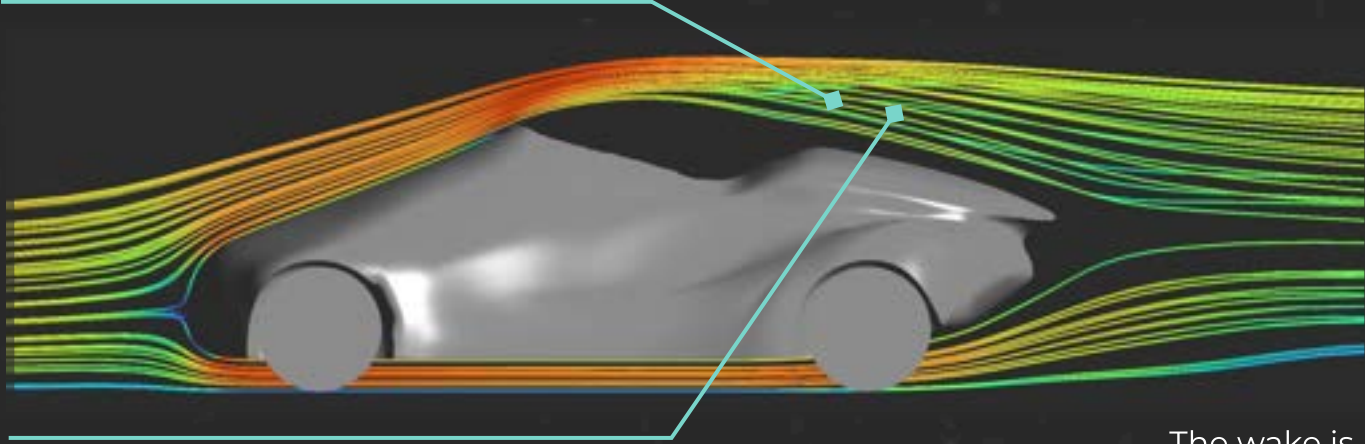
Separation occurs **further toward the back** of the vehicle, showing **delayed detachment** compared to a short flat rear deck.

STREAMLINES & TURBULENCE ANALYSIS

Body Shape Streamlines

Air introduced along the mid-plane

Reattachment occurs **along the upper surface** near the rear hump, with the flow following the surface in a **stable manner**.



The flow remains **relatively organised** as it moves toward the rear, rather than breaking down immediately after the cabin.

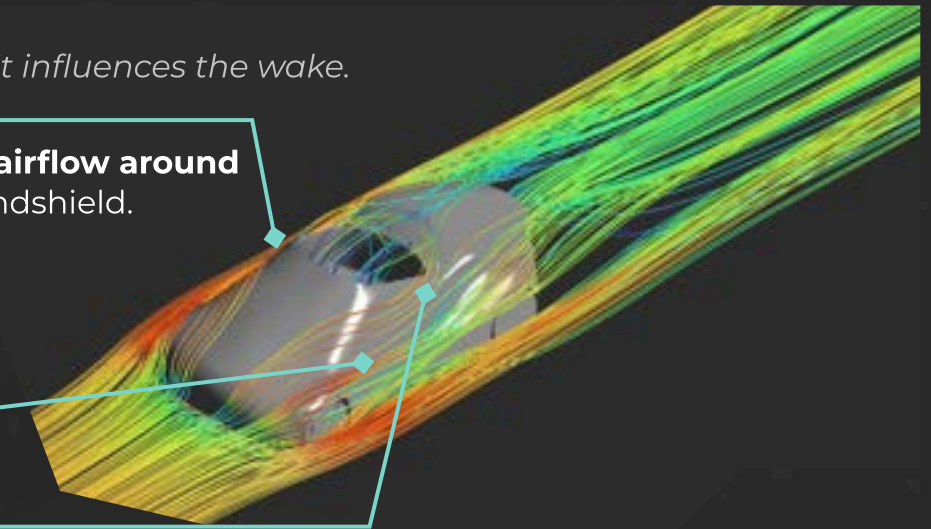
The wake is narrow and gradual.

Air is introduced from a low inlet height

To isolate the side flow behaviour and how it influences the wake.

The body **effectively** directs much of the **airflow around the sides** rather than straight over the windshield.

A lot of the flow **follows the side** and **continues rearward** in an **organised** way, supporting guidance toward the rear channels.



Some **side flow still enters** the **open cabin**, **circulates inside**, and travels along the **underside** of the **windshield** before being **ejected over the rear**.

Instead of reattaching to the rear surface, it **passes above** it, **increasing wake height**.

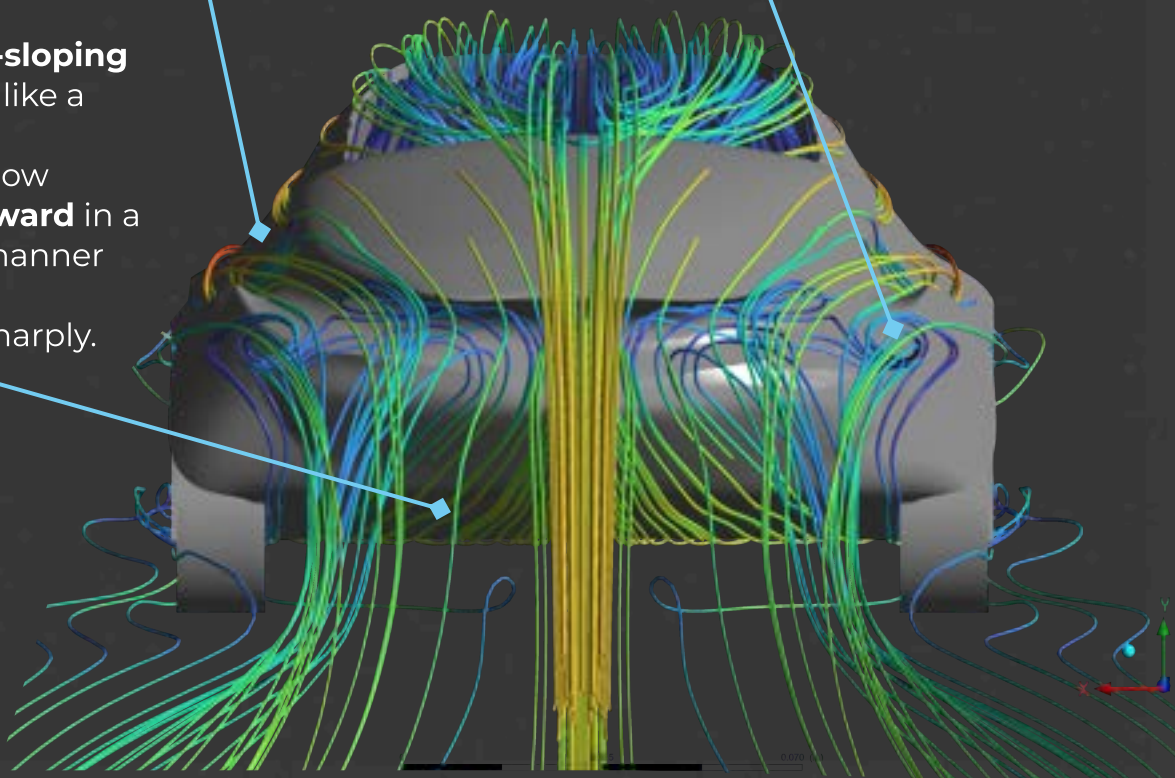
Rear Streamlines

Air introduced around the car

The **air channels successfully direct flow** toward the rear rather than allowing it to detach abruptly at the side edges.

The **upward-sloping bottom** acts like a diffuser. The underbody flow **expands upward** in a **controlled** manner rather than separating sharply.

Vortices form along the **outer rear edges**. These **side vortices add to wake growth** and it means the **flow is still rolling up** as it leaves the body.



Eddy Dissipation (ϵ)

Eddy dissipation is the rate at which **turbulent kinetic energy** is **lost** in the flow due to **viscous effects**. It describes **how quickly** turbulence breaks down and its energy is converted into **heat** at very small scales. [21]

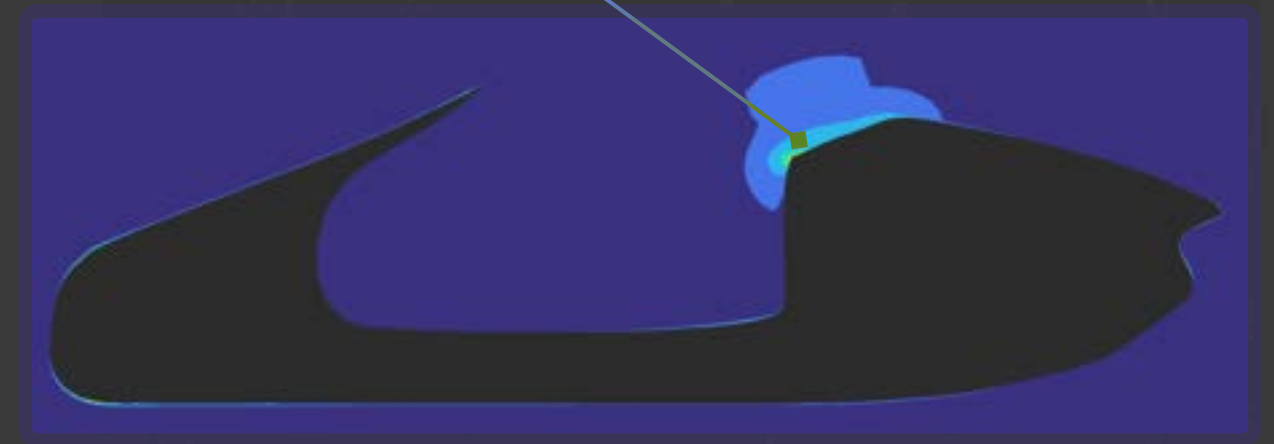
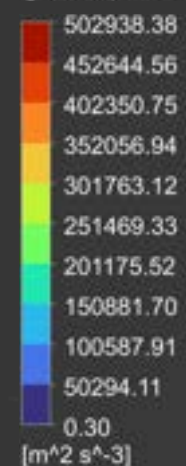
High eddy dissipation is typically found in areas where **turbulence** and **energy loss** are **strongest**, such as:

- *Flow separation*
- *Shear layers*
- *Recirculation regions*
- *Developing wake*

Which all contribute to aerodynamic drag

Turbulent airflow is generated as the flow passes over the cabin and **adjusts to the rear hump**, indicating **increased mixing and shear** between the cabin **outflow** and the **external stream**. This promotes **earlier separation**, enlarges the **wake**, and increases pressure drag.

Turbulence Eddy Dissipation

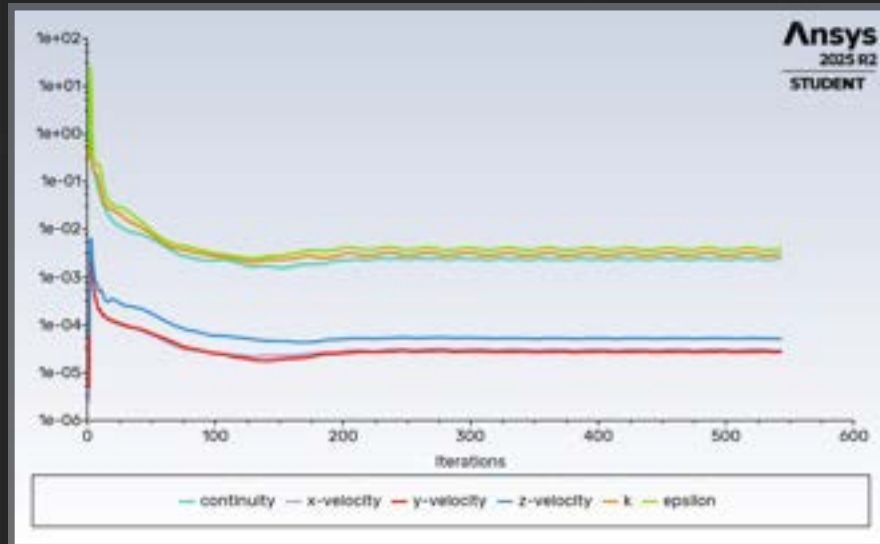


AERODYNAMIC FORCES

[22][23]

Residual Plot

The residual plot shows how the **error in each equation reduces** as the simulation runs. When the residuals level off and **remain stable**, it indicates that the solution has **converged** and is **no longer changing significantly**.



The final residual values were:

- **Continuity:** 2×10^{-3}
- **k:** 3×10^{-3}
- **Epsilon (ϵ):** 4×10^{-3}
- **X-velocity:** 3×10^{-5}
- **Y-velocity:** 3×10^{-5}
- **Z-velocity:** 5×10^{-5}

The velocity residuals reduced to around 10^{-5} , which shows that the momentum equations have been **solved to a high level of accuracy**.

The continuity and turbulence (k & ϵ) residuals stabilised around 10^{-3} .

These values are commonly accepted as **small** for steady external flow simulations, provided the **drag** and **lift** forces have **also stabilised**.

The residuals **dropped rapidly** at the beginning of the simulation and then levelled off.

This shows that the **solution converged** and **remained stable**. Since the drag and lift coefficients also stopped changing with further iterations, the **results can be considered reliable**.

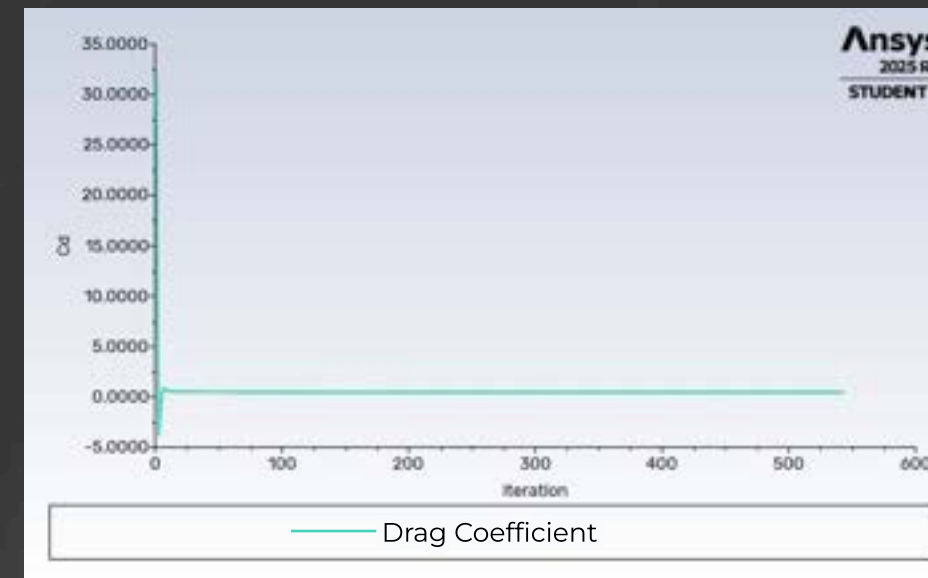
The simulation was carried out using a **symmetry plane**, so only **half** of the **1:17** scale vehicle was modelled to **reduce computational cost**. The **reference area** in Fluent was set to **half** of the model's frontal area to match this setup.

Drag and lift coefficients are calculated by dividing the aerodynamic force by the dynamic pressure and the chosen reference area:

$$C_d = \frac{F_D}{\frac{1}{2} \rho v^2 A} \quad C_l = \frac{F_L}{\frac{1}{2} \rho v^2 A}$$

Since both the force and reference area were **halved**, the coefficients are **unaffected** and represent the **full vehicle behaviour**.

Drag Force



Final Value: (C_d) Drag Coefficient: 0.458

Drag Force is found to be:

$$F_D = \frac{1}{2} C_d \rho A v^2 \quad [29]$$

$$F_D = \frac{1}{2} \times (0.458) \times (1.225) \times (2.904) \times (30.5)^2$$

$$F_D = 758N$$

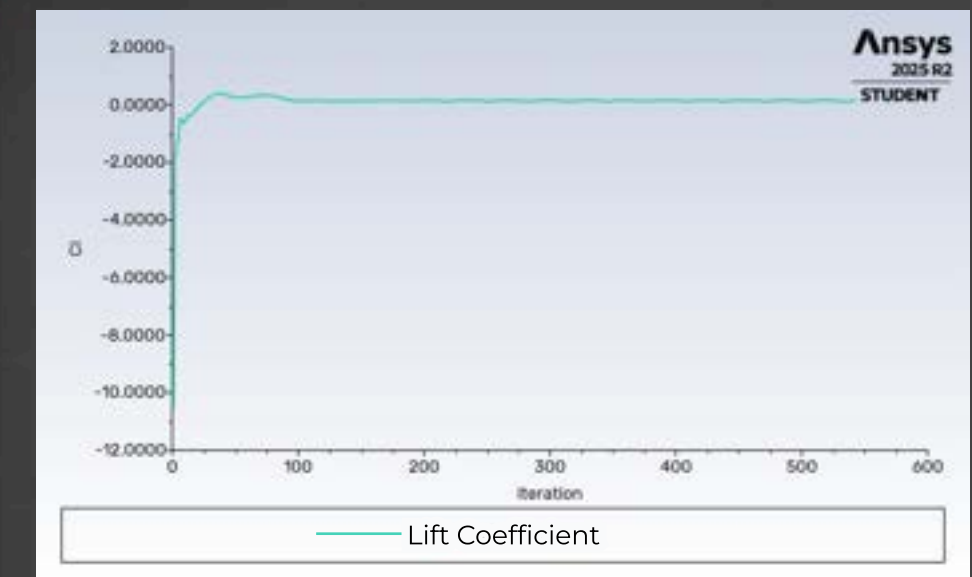
This would be the aerodynamic drag force acting on the full-scale vehicle at 68 mph.



[19]

The drag coefficient is **comparable to a Mercedes-Benz SL600 (0.45)**, meaning the performance is **realistic** for a cabriolet.

Lift Force



Final Value: (C_l) Lift Coefficient: 0.135

Lift Force is found to be:

$$F_L = \frac{1}{2} C_l \rho A v^2 \quad [29]$$

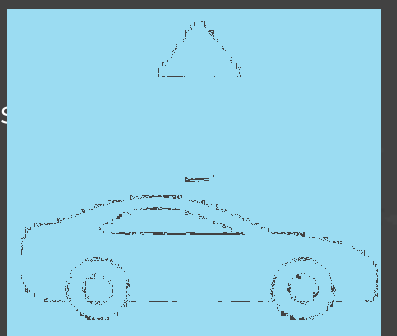
$$F_L = \frac{1}{2} \times (0.135) \times (1.225) \times (2.904) \times (30.5)^2$$

$$F_L = 223N$$

This **positive** value indicates the vehicle generates **upward lift** at this speed.

This corresponds to approximately **23 kg** of reduced vertical load at **68 mph**.

A lift coefficient of **0.135** is **typical** of standard passenger vehicles without downforce-generating aerodynamic devices and **lies within the common range** of **0.10** to **0.20** reported in automotive wind tunnel studies. [28]



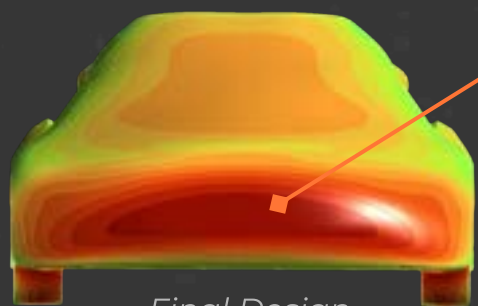
Discussion

The CFD results show that the design achieved a realistic aerodynamic performance for a four-seat cabriolet.

A drag coefficient of **$C_d = 0.458$** is similar to that of the Mercedes-Benz SL600 (**$C_d = 0.45$**)



[19]

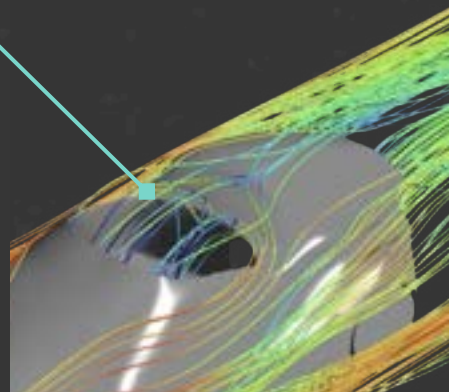


Final Design

This value is **higher** than desired. The elevated drag is likely driven by the **large high-pressure stagnation region** at the nose, where incoming flow is brought close to rest rather than being smoothly redirected around the body.

At the **rear**, **separation** occurs **further downstream** rather than immediately after the cabin. The **pressure reduction** is more **gradual**, with **minimal strong low-pressure suction** directly behind the rear deck. This indicates improved pressure recovery. The **rear shaping** strategy is **effective**, as the flow remains coherent for longer before separating.

Within the cabin region, **separation** occurs at the **windshield header** as expected. The CFD shows **lateral inflow** into the cabin that is **accelerated** over the **rear surface**, **disrupting** the intended **flow path** and reducing the effectiveness of the hump. **Cabin flow** management was therefore only **partially achieved**.



The lift coefficient is positive (**$C_l = +0.135$**). Given the **overall size** and likely **mass** of the vehicle, this level of lift is **unlikely** to have a **significant practical impact** on stability under normal road conditions.

$$F_L = 223N$$

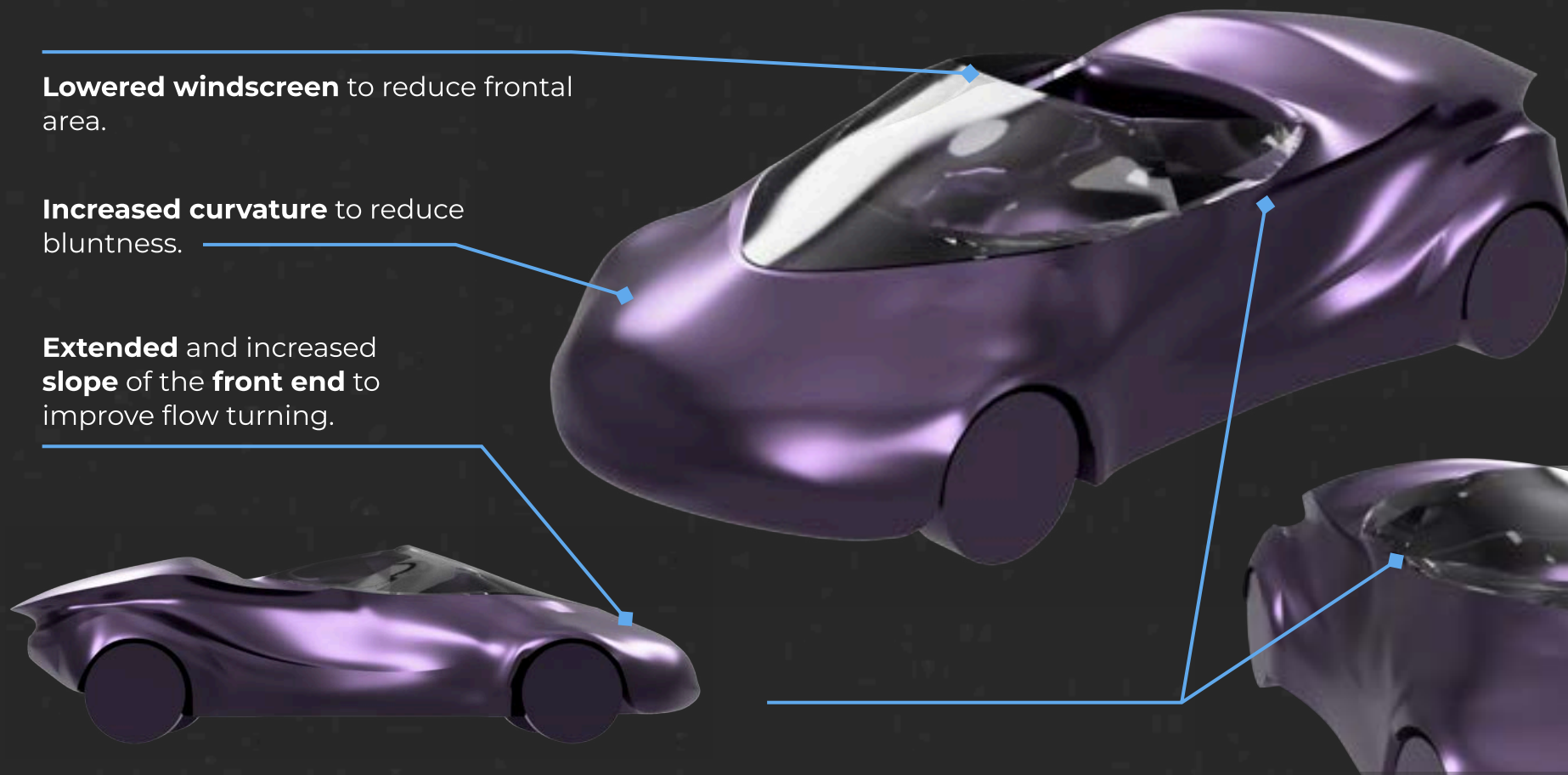
Improved Design

The revised design focuses on **reducing front pressure drag** and **limiting side-entry flow** into the cabin. The **nose** was reshaped to **reduce the high-pressure stagnation** region, while the side and windscreen geometry were refined to **guide airflow over the body** rather than into the cabin.

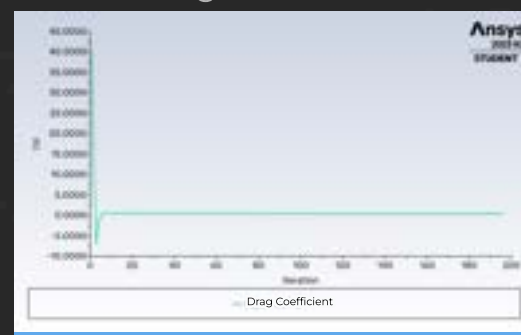
Lowered windscreen to reduce frontal area.

Increased curvature to reduce bluntness.

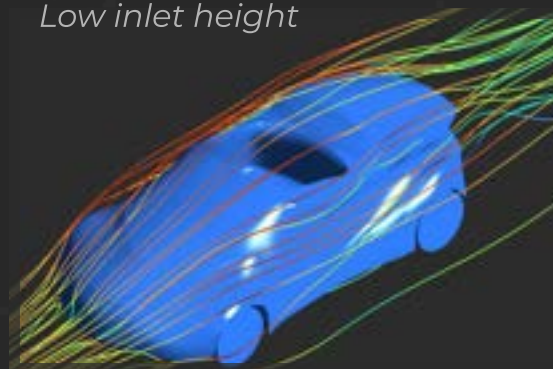
Extended and increased **slope** of the **front end** to improve flow turning.



Drag Coefficient



Low inlet height



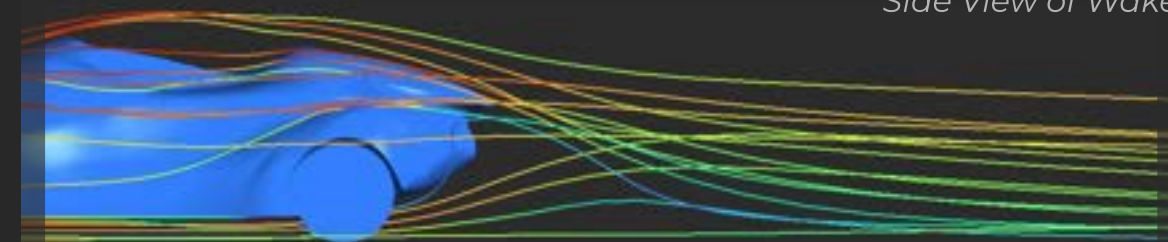
Drag coefficient reduced to **$C_d = 0.405$** (12% reduction).

Frontal area reduced to **2.83 m^2** (2.5% reduction).

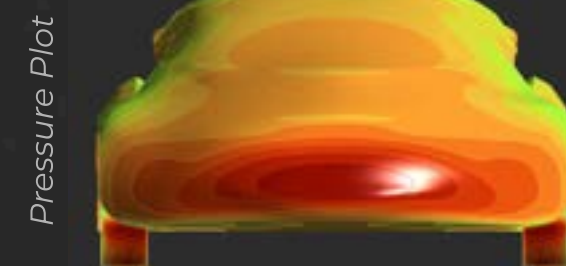
$$F_D = 653N$$

Drag force reduced to **653 N** (14% reduction).

Side inlet flow issue removed therefore the **wake height is reduced** when compared to original design.



New Design



High-pressure region at nose is **smaller** and **more evenly distributed**.

Side View of Wake

References

1. Convertible Aerodynamics [Internet]. SERSOL. 2017. Available from: <https://sersol.weebly.com/workshop/convertible-aerodynamics>

2. Horatiu Boeriu. BMW 4 SERIES CONVERTIBLE - World Premiere [Internet]. BMW BLOG. 2013 [cited 2026 Feb 2]. Available from: <https://www.bmwblog.com/2013/10/12/bmw-4-series-convertible-world-premiere/>

3. AutoSpeed - Aero Basics [Internet]. Autospeed.com. 2020. Available from: https://www.autospeed.com/cms/a_107773/article.html

4. Lei Y. Aerodynamic Designs of Front and Rear End Areas of Cars. Highlights in Science, Engineering and Technology. 2024 Dec 25;120:351–6.

5. BMW 420i Cabrio [Internet]. Vwww.bmw.cc. 2024. Available from: <https://www.bmw.cc/en/all-models/4-series/4-series-cabrio/bmw-4-series-convertible-technical-data.html/bmw-420i-cabrio.bmw>

6. Official Media Website - Mercedes-Benz Cars UK [Internet]. Mercedes-benz-media.co.uk. 2026 [cited 2026 Feb 2]. Available from: <https://mercedes-benz-media.co.uk/models/all-models-cabriolets-c-class-cabriolet>

7. Audi A5 Cabriolet | A5 Range | Audi UK [Internet]. Audi.co.uk. 2022 [cited 2026 Feb 2]. Available from: <https://www.audi.co.uk/en/models/a5/a5-cabriolet/>

8. Ferrari Portofino UK review: Italy's super-GT tested [Internet]. Top Gear. 2019 [cited 2026 Feb 2]. Available from: <https://www.topgear.com/car-reviews/ferrari/qt-2dr-auto/first-drive>

9. Ferrari Portofino - Ferrari.com [Internet]. www.ferrari.com. Available from: <https://www.ferrari.com/en-EN/auto/ferrari-portofino>

10. katz joseph. Aerodynamics of Race Cars [Internet]. ResearchGate. 2006 [cited 2026 Feb 2]. Available from: https://www.researchgate.net/publication/228616843_Aerodynamics_of_race_cars

11. New 2022 BMW 4 Series Convertible - SanDiego.CarPrices.com [Internet]. Autofusion.com. 2022 [cited 2026 Feb 2]. Available from: https://www.autofusion.com/carresearch/TrimsBuild/make_BMW/year_2022/model_4%20Series%20Convertible/confid_sandiegocarprices/

12. Porsche 911 model overview - Porsche AG [Internet]. Porsche AG - Dr. Ing. h.c. F. Porsche AG. Available from: <https://www.porsche.com/international/models/911/>

13. Robinson M. BMW i8 Roadster Review: Take Away The Roof, And The Desire Grows [Internet]. CarThrottle. 2019 [cited 2026 Feb 2]. Available from: <https://www.carthrottle.com/reviews/bmw/bmw-i8-roadster-review-take-away-roof-and-desire-grows>

14. BMW i8 Roadster Price 2026 in Malaysia, Latest Variants Specs&Images [Internet]. PCauto. 2026 [cited 2026 Feb 2]. Available from: <https://www.pcauto.com/my/cars/bmw/i8-roadster>

15. Toyota. Toyota Prius [Internet]. Toyota.com. 2024. Available from: <https://www.toyota.com/prius/>

16. The Aerodynamic Marvel of the Toyota Prius: Understanding Its Low Drag Coefficient [Internet]. oreate.com. 2026 [cited 2026 Feb 5]. Available from: <https://www.oreateai.com/blog/the-aerodynamic-marvel-of-the-toyota-prius-understanding-its-low-drag-coefficient/ab7799db82881cb5b3798bdee4e936be>

17. Monika K, Datta SP. Comparative assessment among several channel designs with constant volume for cooling of pouch-type battery module. Energy Conversion and Management [Internet]. 2022 Jan 1 [cited 2023 Mar 24];251:114936. Available from: <https://www.sciencedirect.com/science/article/pii/S0196890421011122>

18. Wang Z, Guan Q, Zhang H, Wu X, Yu D. Optimized design of liquid-cooled plate structure for flying car power battery system. Journal of Energy Storage. 2024 Sep 1;97:112720–0.

19. Chin C. This Barely Used 1,100-Mile Mercedes-Benz SL600 Is A Rad Time Capsule [Internet]. TopSpeed. 2025 [cited 2026 Feb 19]. Available from: <https://www.topspeed.com/1100-mile-mercedes-benz-sl-600-time-capsule/>

20. Dyson School of Design Engineering, Imperial College London. Thermofluids Module Handbook. Dr Kamyar Hazeri; 2026.

21. Li X, Chen Y, Wang P. Further Development of Eddy Dissipation Model for Turbulent Non-Premixed Combustion Simulation. Energies. 2023 Jun 29;16(13):5043.

22. DE2 Thermo-Fluids: Fluid Mechanics 2022 Notes. Imperial College London; 2022.

23. Dyson School of Design Engineering, Imperial College London. DE2 Thermo-Fluids: Drag – Energy and Design 2022 Notes. Imperial College London; 2022.

24. ANSYS. ANSYS Fluent Software | CFD Simulation [Internet]. Ansys.com. 2017. Available from: <https://www.ansys.com/products/fluids/ansys-fluent>

25. Akkuş F, Işık MZ. A review of thermal management systems of lithium-ion batteries used in electric vehicles. Journal of Traffic and Transportation Engineering (English Edition). 2025 Dec;12(6):1763–83.

26. Ma S, Jiang M, Tao P, Song C, Wu J, Wang J, et al. Temperature effect and thermal impact in lithium-ion batteries: A review. Progress in Natural Science: Materials International [Internet]. 2018 Dec;28(6):653–66. Available from: <https://www.sciencedirect.com/science/article/pii/S1002007118307536>

27. Li S, Zhang C, Zhao Y, Offer GJ, Marinescu M. Effect of thermal gradients on inhomogeneous degradation in lithium-ion batteries. Communications Engineering. 2023 Oct 21;2(1).

28. Byrne CEI. Aerodynamics of Road Vehicles — 4th edition. Edited by W-H. Hucho. SAE International, Warrendale, PA, USA. Materials Park, OH 44073-0002, USA 1998. 918 pp. Illustrated. £78. The Aeronautical Journal. 1999 Aug;103(1026):398.

29. Engineering Toolbox. Drag Coefficient [Internet]. Engineering Toolbox. 2004 [cited 2026 Feb 19]. Available from: https://www.engineeringtoolbox.com/drag-coefficient-d_627.html

30. Automotive Class. How Does an EV Keep Its Battery Cool [Internet]. YouTube. 2025 [cited 2026 Feb 19]. Available from: <https://www.youtube.com/watch?v=a36oYVzG-Is>

31. Manansala J. EV Battery Cooling - How Does It Work? [Internet]. Lectron EV. 2023 [cited 2026 Feb 19]. Available from: https://ev-lectron.com/blogs/blog/ev-battery-cooling-how-does-it-work?srsltid=AfmBOopGE86QL-qFlbgGQQkdetzo8qRSb_kZGUzay_yZ9Hpr49N2Rfl

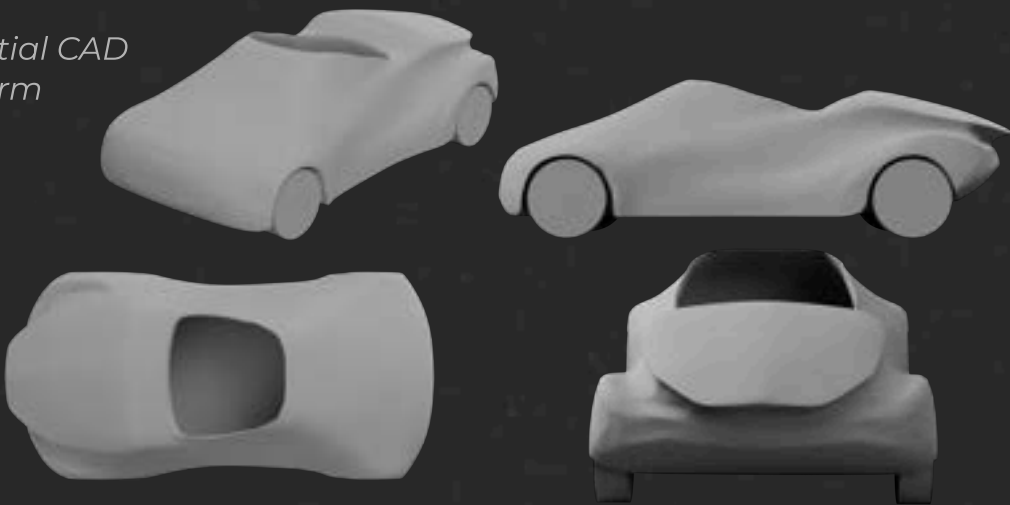
32. BERGMAN TL, LAVINE AS. Fundamentals of Heat and Mass Transfer. 2011.

33. Li X, Wang S. Flow field and pressure loss analysis of junction and its structure optimization of aircraft hydraulic pipe system. Chinese Journal of Aeronautics. 2013 Aug;26(4):1080–92.

34. Lalerz. convection heat transfer convection heat [Internet]. Slideshare. 2025 [cited 2026 Feb 19]. Available from: <https://www.slideshare.net/slideshow/convection-heat-transfer-convection-heat/265011086>

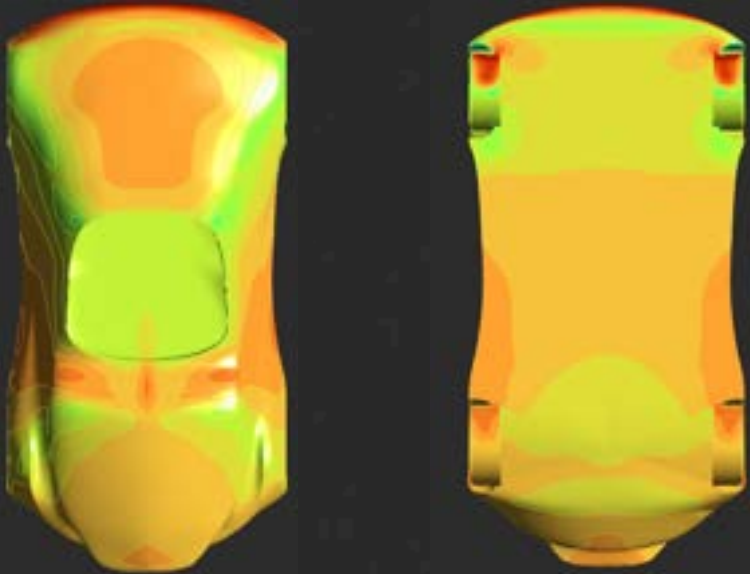
35. Young C. Darcy-Weisbach Equation for Calculating Pressure Losses [Internet]. EngineerExcel. 2022 [cited 2026 Feb 19]. Available from: <https://engineerexcel.com/darcy-weisbach-equation-for-calculating-pressure-losses/>

Initial CAD Form



Pressure Plot Top View

Pressure Plot Bottom View



Sizing the Final Design Sketch for CAD

